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Disorders of Calcium, Phosphate, and Magnesium Metabolism

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CALCIUM HOMEOSTASIS AND DISORDERS OF CALCIUM METABOLISM

Distribution of Calcium in the Organism

Most (>99%) calcium in the body is bound and associated with bony structures. Free calcium, in diffusible nonionized or ionized forms, is found in the intracellular fluid (ICF) and extracellular fluid (ECF) compartments. Similar to potassium, there is a steep concentration gradient between calcium (Ca^{2+}) inside cells and the extracellular compartment (Fig. 11.1).

The serum calcium concentration is tightly regulated within a narrow range by parathyroid hormone (PTH) and calcitriol (1,25-dihydroxycholecalciferol). Fig. 11.2 demonstrates physiologic defense mechanisms used to counter changes in the serum Ca^{2+} concentration.[1] Acute alkalosis causes a decrease in serum Ca^{2+}, and acute acidosis causes an increase in Ca^{2+}. The long-term maintenance of calcium homeostasis depends on intestinal Ca^{2+} absorption, bone accretion and resorption, and urinary excretion (Fig. 11.3).

The extracellular ionized Ca^{2+} concentration is sensed by principal cells in the parathyroid glands via the calcium-sensing receptor (CaSR), which is the primary determinant of PTH synthesis and release. PTH secretion increases or decreases within seconds in response to changes in extracellular ionized Ca^{2+}. The CaSR is present in numerous tissues, including the kidneys.[2] Mutations of the *CASR* gene cause clinical syndromes characterized by hypercalcemia or hypocalcemia (see later).

Intestinal, Skeletal, and Kidney Handling of Calcium

Only about 25% of dietary calcium is absorbed in the gastrointestinal (GI) tract under normal conditions. Intestinal calcium absorption falls precipitously in chronic kidney disease (CKD) because of decreased kidney synthesis of calcitriol. Calcium absorption can be subdivided into transcellular and paracellular flow (Fig. 11.4). Transcellular flux takes place through the apical transient receptor potential channel vanilloid subtype 6 (TRPV6) calcium channel.[3] Calcitriol stimulates active calcium transport by binding to and activating the vitamin D receptor (VDR), which induces expression of TRPV6, calbindin-D_{9k}, and Ca^{2+}-ATPase (PMCA1b).[4] Other hormones, including estrogens, prolactin, growth hormone, and parathyroid hormone (PTH), as well as the amount of dietary calcium intake, also stimulate intestinal calcium absorption, either directly or indirectly (Fig. 11.5).[5]

The primary source of vitamin D derives from exposure to ultraviolet light, which converts 7-dehydrocholesterol to cholecalciferol (vitamin D3) in the skin. Cholecalciferol can also be obtained through supplementation and dietary sources; however, the cholecalciferol content of most foods is low. Ergocalciferol (vitamin D_2) is an alternative vitamin D supplement that does not occur naturally in humans. Cholecalciferol and ergocalciferol (which contains an additional methyl group) have minimal inherent biologic function and require two hydroxylation steps for full hormonal activity. The first step, 25-hydroxylation, occurs in the liver by vitamin D 25-hydroxylase encoded by the *CYP2R1* gene. This step is believed to be nonrate limiting; however, polymorphisms in *CYP2R1* are associated with differences in circulating 25-hydroxyvitamin D concentrations in the general population. The serum 25-hydroxyvitamin D concentration is widely accepted as the measure of vitamin D stores. Further hydroxylation to 1,25-dihydroxyvitamin D (calcitriol) is performed by 25-hydroxyvitamin D-1 alpha hydroxylase encoded by the *CYP27B1* gene. This step occurs predominantly in the kidneys but can also occur in macrophages, parathyroid glands, and other tissues. The activated, 1-hydoxy forms of vitamin D are subsequently catabolized by the CYP24A1 enzyme (cytochrome P450 family 24 subfamily A member 1) to prevent toxicity. Serum 1,25-dihydroxyvitamin D concentrations are approximately 1000-times lower than 25-hydroxyvitamin D, highlighting the tight hormonal control of the final hydroxylation step.

An increase in intestinal calcium absorption is required in puberty and pregnancy, whereas in lactation, calcium is mostly obtained from maternal bone.[6] In puberty and pregnancy, calcitriol synthesis is increased to enhance calcium absorption. Intestinal calcium absorption is also increased in states of vitamin D excess and acromegaly. Rarely, the ingestion of calcium and alkali in large quantities can overwhelm checks on calcium absorption, resulting in hypercalcemia (milk-alkali syndrome). A decrease in intestinal calcium transport occurs in advanced age, kidney disease (because of reduced calcitriol synthesis), gastrectomy, intestinal malabsorption syndromes, diabetes mellitus, corticosteroid treatment, and estrogen deficiency and with dietary factors such as high vegetable fiber and fat content, low calcium-to-phosphate ratio in food, and fructose ingestion. The decrease in calcium absorption in older adults probably results from multiple factors in addition to lower serum calcitriol and intestinal VDR.[7]

The net balance between calcium entry and exit is positive during skeletal growth in children, zero in young adults, and negative in older persons. Exchangeable skeletal calcium contributes to the maintenance of extracellular calcium homeostasis. Several growth factors, hormones, and genetic factors participate in the differentiation from mesenchymal precursor cell to osteoblast and maturation of the osteoclast from its granulocyte-macrophage precursor cell (Fig. 11.6). Bone formation and resorption is also regulated by many hormones, growth factors, and mechanical factors[8,9] (Fig. 11.7).

The kidneys regulate minute-by-minute calcium balance, and the intestine, kidney, and skeleton ensure homeostasis in the mid and long term (Fig. 11.8).[10] The synthesis of calcitriol from substrate 25-hydroxyvitamin D is increased by PTH and reduced by fibroblast growth factor 23 (FGF-23) and perhaps hyperphosphatemia. The adjustment of blood calcium is mainly achieved by modulation of tubular calcium reabsorption, perfectly compensating minor increases or decreases in the filtered calcium load (normally about 220 mmol

[8800 mg] per day; see Fig. 11.3). The bulk of calcium reabsorption in the kidneys occurs in the proximal tubules following the convective flow of salt and water, with relatively smaller contributions from the thick ascending limb and distal tubular segments.

Excess volume delivery to the kidneys, such as with a high-sodium diet, may diminish the concentration gradient between the proximal tubule and peritubular capillary, reducing the driving force for calcium absorption and increasing urinary calcium excretion. This mechanism is suspected to contribute to the pathogenesis of calcium-based kidney stones and provides a rationale for recommending a low-sodium diet in this setting. On the other hand, volume depletion increases the reabsorption of salt, water, and, by convection, calcium in the proximal tubules, exacerbating states of calcium excess. For this reason, intravascular volume repletion is an essential first step in the treatment of hypercalcemia.

In the thick ascending limb (TAL), calcium transport is paracellular, functionally linked with activity of the Na^+-K^+-$2Cl^-$ transporter and the attendant lumen-positive transepithelial voltage. Stimulation of the basolateral CaSR decreases the paracellular pathway permeability to calcium.[11] The result is decreased calcium reabsorption through the Claudin-16 and 19 complex in tight junctions (Fig. 11.9). In contrast, understimulation of the CaSR by low serum calcium levels increases tubular calcium reabsorption. In the distal tubules, active calcium transport occurs via the transcellular route through the transient receptor potential channel vanilloid subtype 5 (TRPV5). This channel is located in the apical membrane and coupled with a specific basolateral calcium-ATPase (PMCa1b) and Na^+-Ca^{2+} exchanger (NCX1). Both PTH and calcitriol increase distal tubular Ca^{2+} transport. Several classes of diuretics alter kidney calcium handling. Loop diuretics increase urinary calcium excretion by inhibiting the Na^+-K^+-$2Cl^-$ cotransporter, whereas thiazide diuretics and amiloride increase distal tubular calcium absorption.

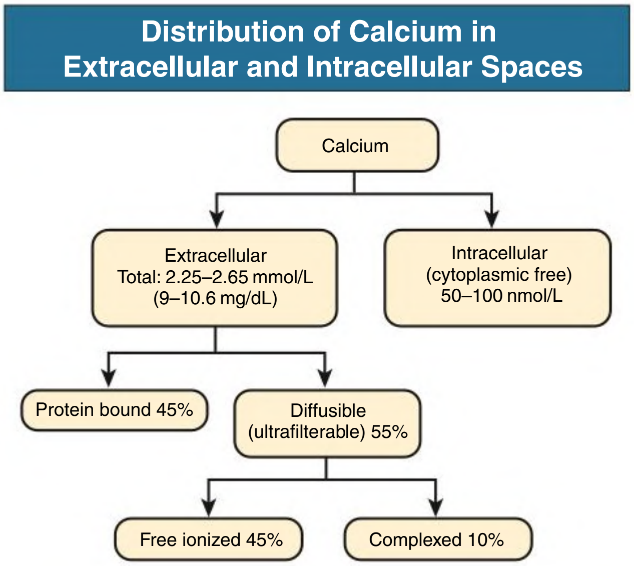

Fig. 11.1 Calcium distribution in extracellular and intracellular spaces.

HYPERCALCEMIA

Routine measurement of the serum calcium concentration [Ca^{2+}] reports the total concentration in the tube, including bound and unbound fractions. Increased serum [Ca^{2+}] can therefore result from an increase in serum proteins (false hypercalcemia) or reflect a real increase in ionized [Ca^{2+}] (true hypercalcemia). If laboratory assays for ionized calcium are not available, serum [Ca^{2+}] can be corrected using serum albumin: each

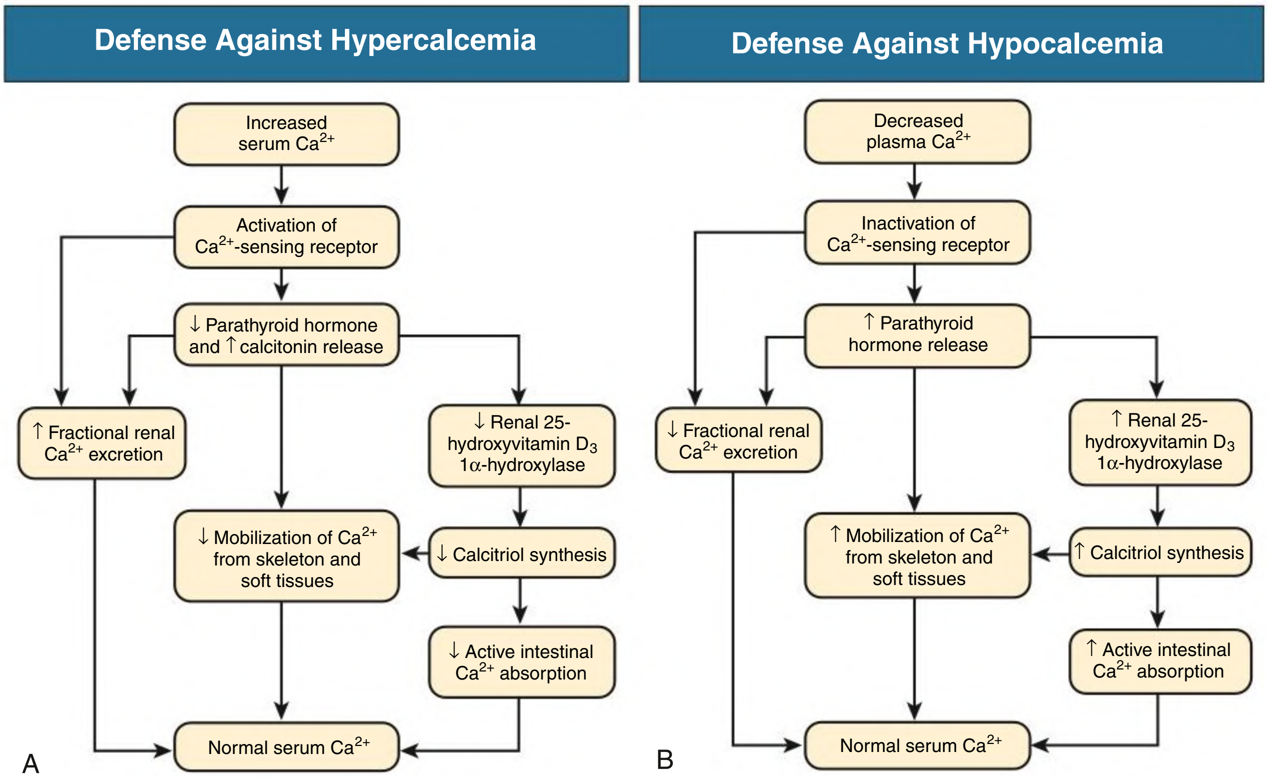

Fig. 11.2 Calcium Regulation. Physiologic defense mechanisms against increases or decreases in serum calcium levels. (A) Hypercalcemia. (B) Hypocalcemia. (Modified from Kumar R. Vitamin D and calcium transport. *Kidney Int*. 1991;40:1177–1189.)

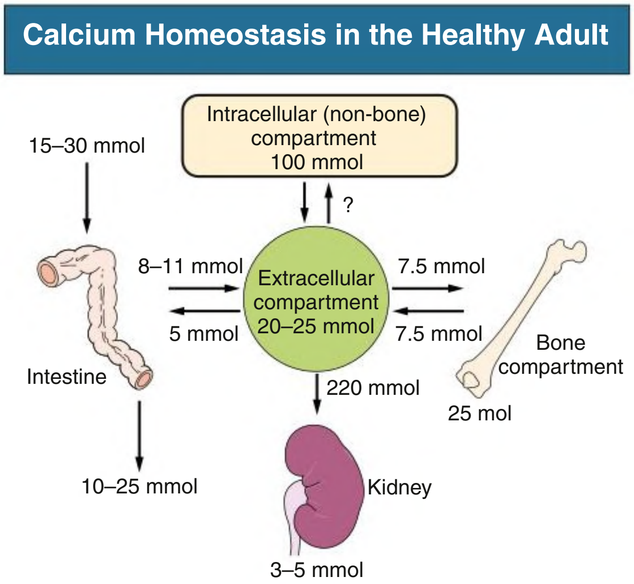

Fig. 11.3 Calcium Homeostasis in Healthy Adults. Net zero Ca^{2+} balance is the result of net intestinal absorption (absorption minus secretion) and urinary excretion, which by definition are the same. After its passage into the extracellular fluid, Ca^{2+} enters the extracellular space, is deposited in bone, or is eliminated via the kidneys. Entry and exit fluxes between the extracellular and intracellular spaces (skeletal and nonskeletal compartments) are also of identical magnitude under steady-state conditions.

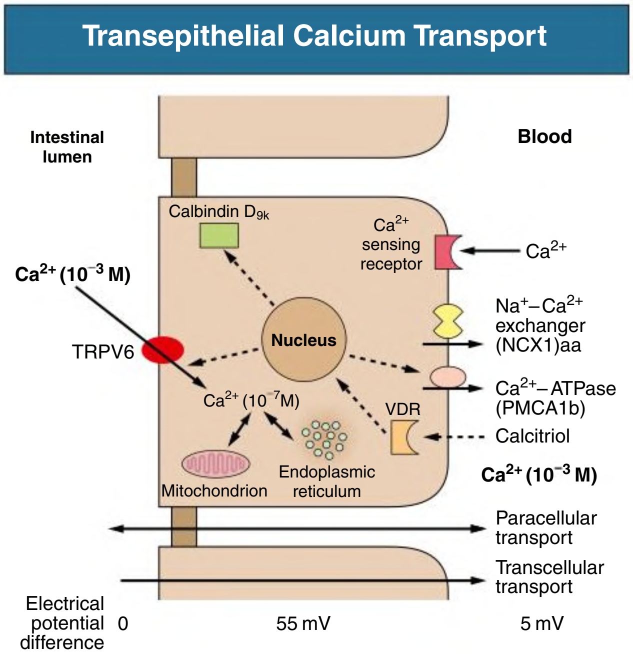

Fig. 11.4 Transepithelial Calcium Transport in the Small Intestine. Calcium penetrates into the enterocyte channels via a transient receptor potential calcium channel (TRPV6) through the brush border membrane along a favorable electrochemical gradient. Under physiologic conditions, the cation is pumped out of the cell at the basolateral side against a steep electrochemical gradient by the adenosine triphosphate–consuming pump Ca^{2+}-ATPase. When there is a major elevation of intracytoplasmic Ca^{2+}, the cation leaves the cell using the Na^{+}-Ca^{2+} exchanger. Passive Ca^{2+} influx and efflux are sensitive to calcitriol, which binds the vitamin D receptor (VDR).

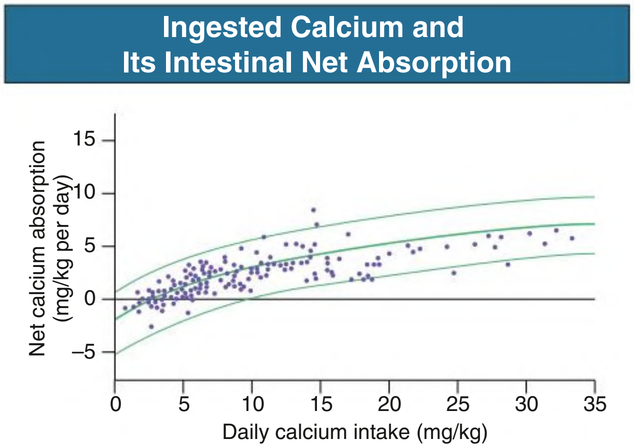

Fig. 11.5 Calcium Ingestion and Absorption. Relationship between ingested calcium and its absorption in the intestinal tract (net) in healthy young adults. (From Wilkinson R. Absorption of calcium, phosphate, and magnesium. In: Nordin BEC, ed. *Calcium and Magnesium Metabolism*. Churchill Livingstone; 1976:36–112.)

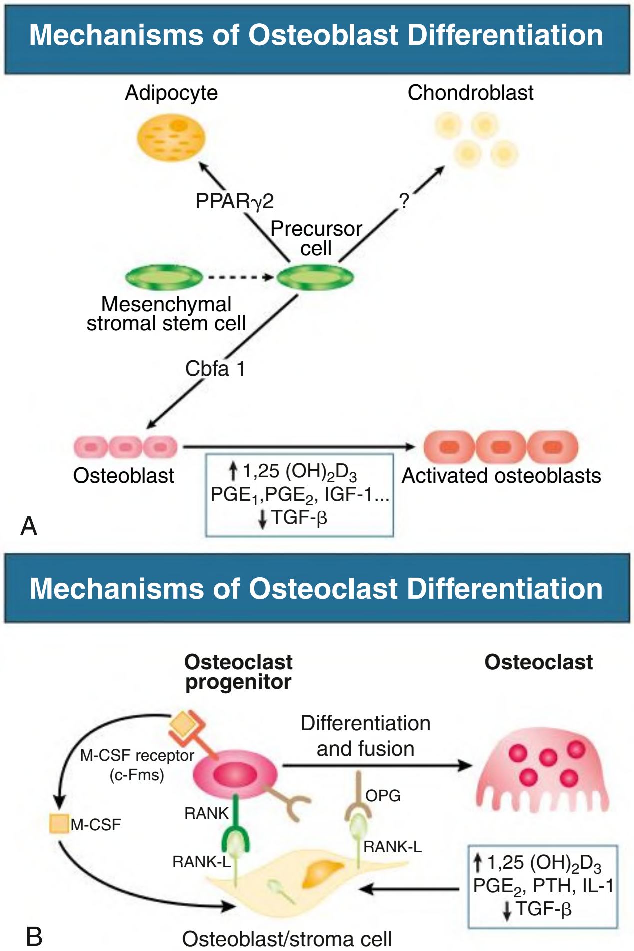

Fig. 11.6 Mechanisms of Osteoblast Differentiation. (A) Major growth factors and hormones controlling the differentiation from the mesenchymal precursor cell to the osteoblast. (B) The major growth factors, cytokines, and hormones controlling osteoblast and osteoclast activity. *IGF,* Insulin-like growth factor; *IL,* interleukin; *M-CSF,* macrophage colony-stimulating factor; *OPG,* osteoprotegerin; *PGE*$_2$, prostaglandin E_2; *PPAR,* peroxisome proliferator–activated receptor; *PTH,* parathyroid hormone; *RANKL,* receptor activator of nuclear factor-κβ ligand; *TGF,* transforming growth factor.

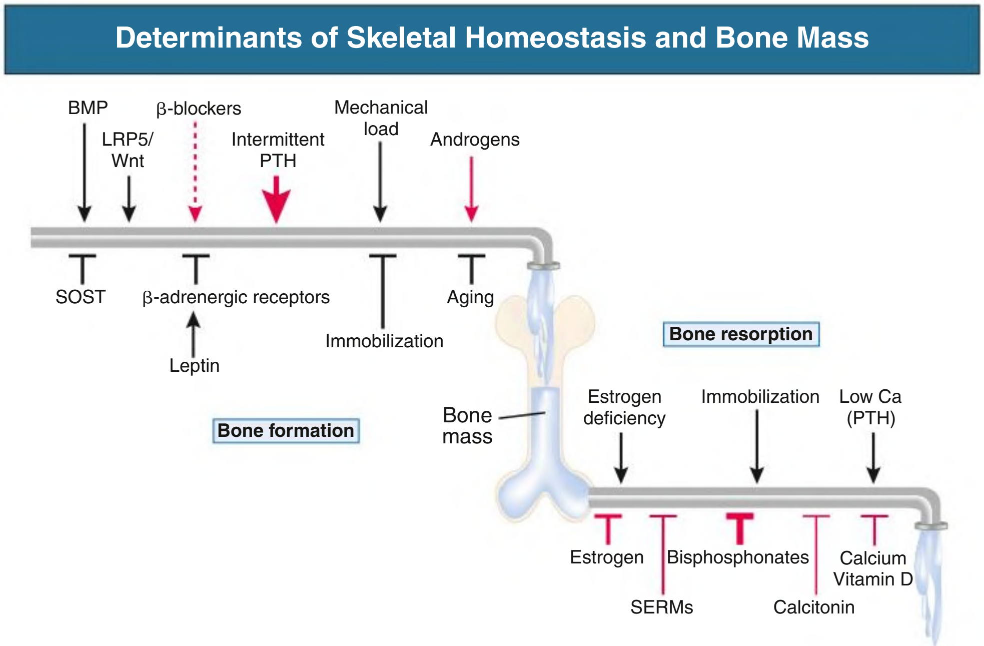

Fig. 11.7 **Determinants of Skeletal Homeostasis and Bone Mass.** Physiologic *(black)* and pharmacologic *(red)* stimulators and inhibitors of bone formation and resorption are listed with the relative impact (represented by thickness of *arrows*). *BMP,* Bone morphogenetic protein; *LRP5,* low-density lipoprotein receptor–related protein 5; *PTH,* parathyroid protein; *SERMs,* selective estrogen receptor modulators; *SOST,* sclerostin. (From Kinyamu HK. Association between intestinal vitamin D receptor, calcium absorption, and serum 1,25 dihydroxyvitamin D in normal young and elderly women. *J Bone Miner Res.* 1997;12:922–928.)

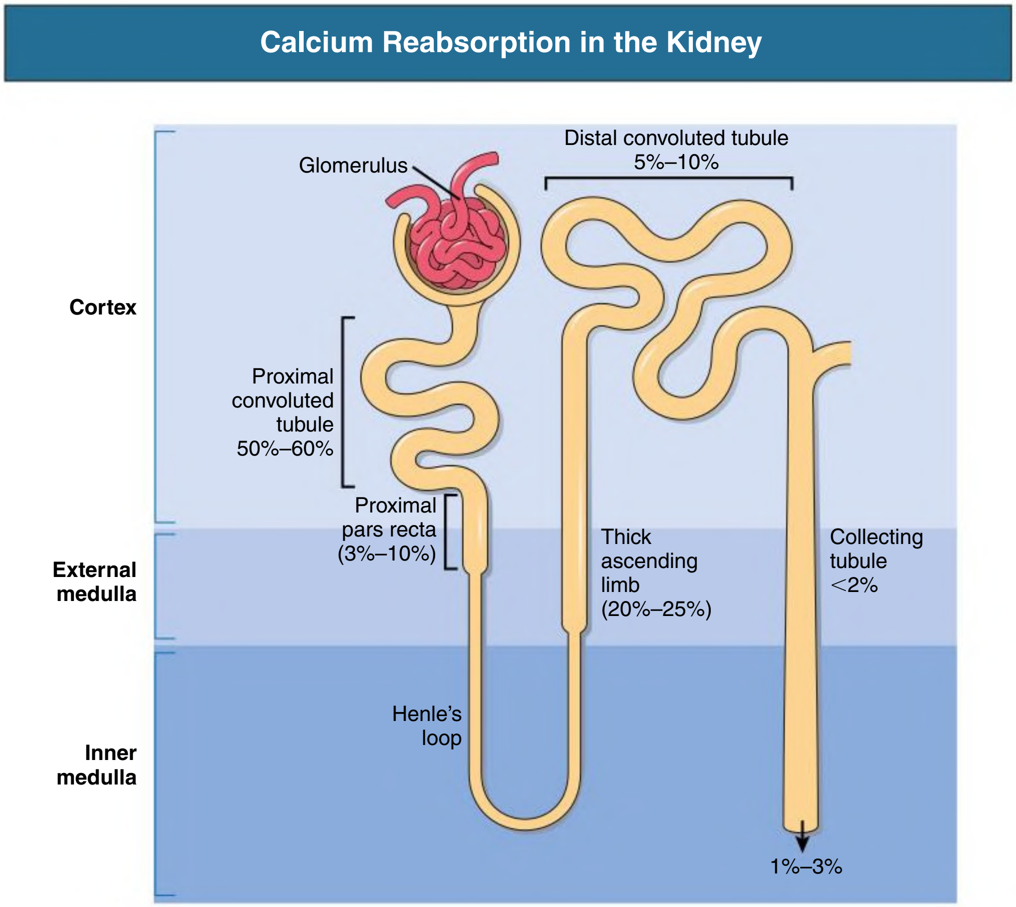

Fig. 11.8 **Sites of Calcium Reabsorption.** Percentage of Ca^{2+} absorbed in various segments of the renal tubule after glomerular ultrafiltration. (Modified from Martin T, Gooi JH, Sims NA. Molecular mechanisms in coupling of bone formation to resorption. *Crit Rev Eukaryot Gene Expr.* 2009[19]:73–88.)

1.0 g/dL higher serum albumin concentration is associated with a concomitant 0.20 to 0.25 mmol/L (0.8–1.0 1.0 mg/dL) greater serum [Ca^{2+}]. However, simple correction for serum albumin may not be valid in patients with CKD stages 3 to 5.[12]

Causes of Hypercalcemia

Hypercalcemia may be caused by an increase in intestinal calcium absorption, excess bone resorption, enhanced kidney tubular reabsorption, or a combination of these processes. The most common clinical causes of hypercalcemia are malignancy and primary hyperparathyroidism, which can be distinguished by measurement of PTH. Regardless of the primary cause, secondary volume depletion prolongs hypercalcemia by interfering with urinary calcium excretion (Fig. 11.10).

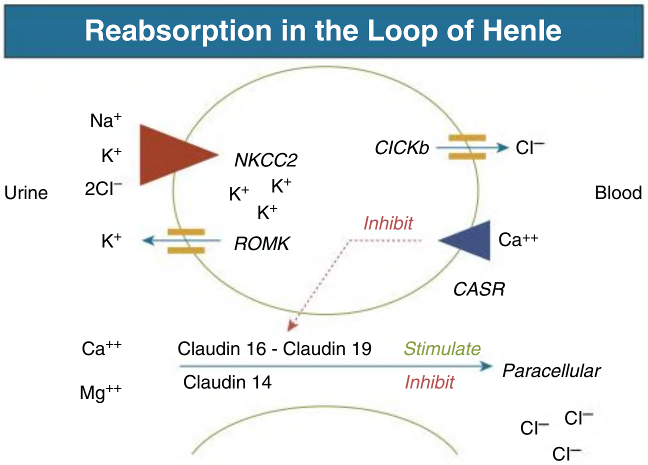

Fig. 11.9 Reabsorption in the Loop of Henle. Transport of Na^+, K^+, $2Cl^-$ into the cell via *NKCC2* drives paracellular reabsorption of calcium and magnesium through a claudin 16-claudin 19 complex. Stimulation of the calcium-sensing receptor on the basolateral cell surface, such as by hypercalcemia, inhibits calcium and magnesium reabsorption.

Malignant Neoplasia

An important cause of hypercalcemia is excessive bone resorption induced by neoplastic processes, usually solid tumors.[13] Tumors of the neck, breast, lung, and kidney are the most common, followed by hematopoietic neoplasias, particularly myeloma. Tumors causing hypercalcemia act on the skeleton by direct invasion (metastases) or by producing factors that stimulate osteoclastic activity (paraneoplastic processes). The most common tumor-derived factor is PTH-related protein (PTHrP). Only 8 of the 13 first amino acids of PTHrP are homologous to the N-terminal fragment of PTH, but the effects of both hormones on target cells are similar because they share a common receptor (the PTH/PTHrP receptor, PTH1R). In multiple myeloma and lymphomas, osteoclast-activating factors secreted by clonal cells include interleukins (IL-1α, IL-1β, IL-6) and tumor necrosis factor (TNF)-α. Certain tumors, especially kidney masses, may secrete other osteoclast-activating factors, such as PGE_1 and PGE_2. Some lymphoid tumors, including Hodgkin disease, T-cell lymphoma, and leiomyoblastoma, can synthesize excess quantities of calcitriol, which promote hypercalcemia.

Primary Hyperparathyroidism

In more than 80% of patients, hyperparathyroidism is caused by an adenoma of a single parathyroid gland; 10% to 15% have diffuse hyperplasia of all glands, and less than 1% have parathyroid carcinoma.

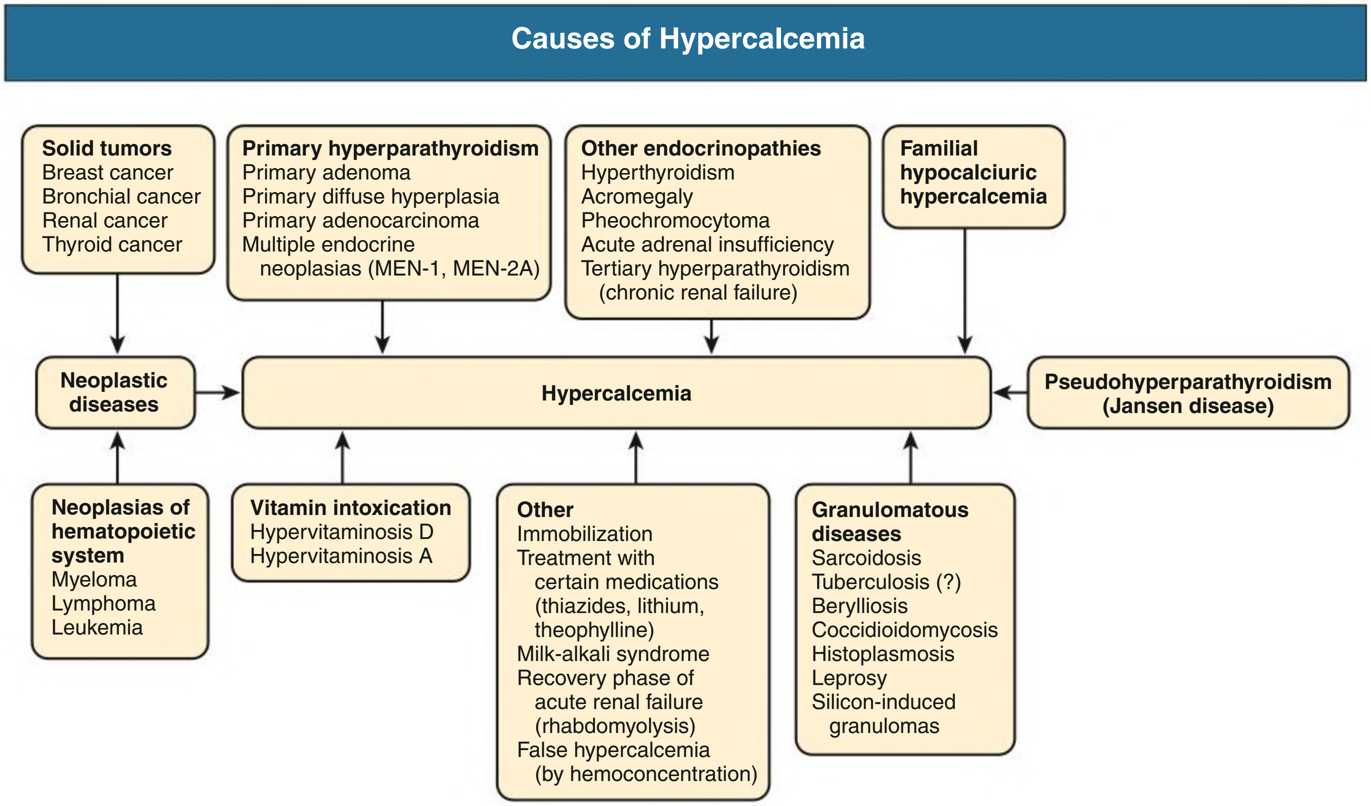

Fig. 11.10 Causes of Hypercalcemia. Neoplastic diseases and primary hyperparathyroidism are the most common causes of hypercalcemia. (From Harada S, et al. Control of osteoblast function and regulation of bone mass. *Nature*. 2003;12:922–928.)

Primary hyperparathyroidism typically presents in middle age with asymptomatic mild hypercalcemia and a high or high normal serum PTH concentration. Under normal conditions, even a mild increase in serum [Ca^{2+}] should potently suppress PTH secretion. Consequently, primary hyperparathyroidism should be suspected when both serum [Ca^{2+}] and PTH are at the higher end of the normal range. Urinary calcium excretion is increased in primary hyperparathyroidism, which may assist in the diagnosis.

Primary hyperparathyroidism is usually idiopathic but can be inherited either as diffuse hyperplasia of the parathyroid glands alone or as a component in multiple glandular hereditary endocrine disorders. Patients with multiple endocrine neoplasia type 1 (MEN-1) have various combinations of parathyroid, anterior pituitary, enteropancreatic, and other endocrine tumors, resulting in hypersecretion of prolactin, gastrin, and PTH. MEN-1 is caused by inactivating germ-line mutations of a tumor suppressor gene *(MEN-1)* that is inherited as an autosomal dominant trait. In MEN-2A, the thyroid medulla and the adrenal medulla are involved, resulting in hypersecretion of calcitonin and catecholamines. MEN-2A is caused by activating mutations of the *RET* proto-oncogene. It is also inherited as an autosomal dominant trait. MEN-4 is characterized by the occurrence of parathyroid and anterior pituitary tumors possibly associated with tumors of the adrenals, kidneys, and reproductive organs; it is because of cyclin-dependent kinase inhibitor (CDNK1B) mutations.[14]

Granulomatous Diseases

The formation of macrophage-laden granulomas in specific infectious and inflammatory diseases can promote unchecked activity of macrophage-associated 1 alpha hydroxylase activity, leading to excess calcitriol synthesis and hypercalcemia. Common granulomatous diseases associated with hypercalcemia are sarcoidosis and tuberculosis; other conditions include histoplasmosis, coccidioidomycosis, berylliosis, and leprosy. Laboratory measurement of the serum 1,25-dihydroxyvitamin D concentration can help identify these conditions.

Other Endocrine Causes

Other endocrine disorders associated with moderate hypercalcemia include hyperthyroidism, acromegaly, pheochromocytoma, and acute adrenal insufficiency.

Inherited Causes

Familial Hypocalciuric Hypercalcemia

Familial hypocalciuric hypercalcemia (FHH) is a rare hereditary disease, most often caused by inactivating mutations in the *CASR* gene with autosomal dominant transmission,[15] more rarely in the gene for G-protein subunit α_{11} (GNA11), or adaptor-related protein complex 2, sigma 1 subunit (AP2S1).[16] The key clinical finding in FHH is an inappropriately low urinary calcium excretion in the setting of high-normal serum [Ca^{2+}]. Serum PTH levels are often normal or moderately increased, potentially mimicking the diagnosis of primary hyperparathyroidism. The key distinction is low urine calcium excretion in FHH, assessed by calculating the fractional excretion of calcium (FE_{Ca}) from a 24-hour urine collection:

$$\text{Fraction excretion of calcium } (FE_{Ca}) = \frac{(\text{Urine calcium}) \times (\text{Serum creatinine})}{(\text{Serum calcium}) \times (\text{Urine creatinine})}$$

In FHH, the FE_{Ca} is usually less than 0.01; however, some overlap may still be seen with primary hyperparathyroidism.[17] FHH rarely leads to severe hypercalcemia except in the neonatal period.

Jansen Disease

Jansen disease is a rare hereditary form of short-limbed dwarfism characterized by severe hypercalcemia, hypophosphatemia, and metaphyseal chondrodysplasia. The condition is caused by activating mutations of the gene coding for the PTH1R receptor, a particular form of pseudohyperparathyroidism.

Other Causes

Hypercalcemia is seen with protracted bedrest, especially in the setting of prolonged hospitalization. Hypercalcemia is rarely observed with intoxication from vitamin A or D supplements and thiazide diuretics. Large doses of oral calcium (5–10 g/day), when ingested with alkali (antacids), can lead to hypercalcemia and nephrocalcinosis (milk-alkali syndrome).

Hypercalcemia in Chronic Kidney Disease

The natural history of CKD includes a gradual reduction in serum [Ca^{2+}] because of reduced calcitriol synthesis. The presence of hypercalcemia in patients with advanced CKD or end-stage kidney disease (ESKD) should alert the clinician to a potential iatrogenic cause, specifically the overadministration of calcium-containing phosphate binders and vitamin D receptor agonists (calcitriol, paricalcitol). A second potential cause of hypercalcemia in ESKD is tertiary hyperparathyroidism, in which the parathyroid glands lose responsiveness to therapy with severe elevation of serum PTH.

Clinical Manifestations

The severity of symptoms and signs caused by hypercalcemia depends not only on the degree but also the rapidity of development. Common symptoms are fatigue, muscle weakness, inability to concentrate, nervousness, increased sleepiness, and depression. GI symptoms include constipation, nausea and vomiting, and, rarely, peptic ulcer disease or pancreatitis. Kidney-related signs include hypertension, polyuria (secondary to nephrogenic diabetes insipidus), salt wasting, kidney stones and their complications, and, occasionally, tubulointerstitial disease with medullary and, to a lesser extent, cortical nephrocalcinosis. Patients with chronic hypercalcemia often present with a modest elevation in the serum creatinine concentration, indicating reduced GFR. Neuropsychiatric manifestations are common and include headache, loss of memory, somnolence, stupor, and, rarely, coma. Ocular symptoms include conjunctivitis from crystal deposition and, rarely, band keratopathy. The electrocardiogram (ECG) may show shortening of the QT interval and coving of the ST wave. Hypercalcemia may increase cardiac contractility and can amplify digitalis toxicity.

Diagnosis

In addition to the clinical history, examination, and review of medications, measurement of ionized [Ca^{2+}] and serum PTH are the first steps in investigating the cause of hypercalcemia. Primary hyperparathyroidism should be suspected if the PTH concentration is high or high-normal in the presence of a high or high-normal serum [Ca^{2+}], although FHH is not excluded by this laboratory pattern. If PTH is appropriately suppressed with hypercalcemia, then the possibility of a neoplastic disorder should be considered. Subsequent laboratory steps to determine the cause of hypercalcemia include serum protein electrophoresis and measurement of PTHrP, 25-hydroxyvitamin D, and 1,25-dihydroxyvitamin D.

Treatment

The treatment of hypercalcemia is aimed at the underlying cause, although severe (>3.25 mmol/L) and symptomatic hypercalcemia always requires rapid correction. Initially, the patient should be volume expanded with isotonic saline or balanced crystalloids to inhibit tubular calcium reabsorption and enhance urinary calcium excretion. Only when euvolemia is established can loop diuretics be considered to further augment urinary calcium excretion; however, intravenous (IV) fluids should be continued to prevent hypovolemia. Volume expansion must be carefully managed in patients with heart failure or advanced kidney disease. Oral intake, IV administration of fluids and electrolytes, and acid-base balance should be carefully monitored.

Bisphosphonates are the treatment of choice, especially in patients with hypercalcemia associated with cancer.[13] These agents inhibit bone resorption and calcitriol synthesis. Bisphosphonates can be administered orally in less severe disease or by IV in severe hypercalcemia. Common bisphosphonates include pamidronate 15 to 90 mg IV over 1 to 3 days and zoledronate 4 mg IV once; IV doses should be infused in 500 mL of isotonic saline or dextrose over at least 2 hours (pamidronate) or 15 minutes (zoledronate). Although package warnings state that bisphosphonates should be used with caution in patients with CKD, this warning pertains to the theoretical possibility of inducing hypocalcemia. Bisphosphonates have been safely used in patients with CKD for the correction of hypercalcemia. Calcitonin acts within hours, especially after IV administration. Human, porcine, or salmon calcitonin can be given. However, calcitonin often has no effect, or only a short-term effect, because of the rapid development of tachyphylaxis.

Denosumab, a monoclonal antibody to the receptor activator of nuclear factor κ-B ligand (RANKL), is a potent inhibitor of bone resorption that can be useful in bisphosphonate-refractory hypercalcemia. Denosumab is not removed by the kidneys. The typical dose is 120 mg subcutaneously and can be repeated no earlier than 1 week after the first administration.[18]

Corticosteroids such as prednisone (or prednisolone), 0.5 to 1.0 mg/kg daily, are mainly indicated in patients with sarcoidosis or tuberculosis, to decrease macrophage synthesis of calcitriol. Corticosteroids also can be used in patients with hypercalcemia associated with some hematopoietic tumors (e.g., myeloma, lymphoma) and even for some solid tumors such as breast cancer. Ketoconazole, an antifungal agent that can inhibit renal and extrarenal calcitriol synthesis, can also be used to treat hypervitaminosis D.

In rare cases of malignant hypercalcemia, treatment with prostaglandin antagonists such as indomethacin or aspirin can be successful. Hyperkalemia and impaired kidney function may occur with indomethacin. Hypercalcemia caused by thyrotoxicosis can rapidly resolve with IV administration of propranolol or less rapidly with oral administration.

In patients with primary hyperparathyroidism, consideration for parathyroidectomy is based on the severity of hypercalcemia and the accompanying complications of the disease, including loss of bone mineral density, nephrolithiasis, and kidney dysfunction. For nonsurgical, asymptomatic candidates, potential medical therapies include bisphosphonates and the CaSR agonist cinacalcet (a calcimimetic).[19] Cinacalcet is also effective in patients with parathyroid carcinoma. Surgical parathyroidectomy remains an important therapeutic option in patients with ESKD who are unresponsive to vitamin D analogs and calcimimetics.[20]

HYPOCALCEMIA

A decrease in total serum [Ca^{2+}] may occur as a result of reduced serum albumin (false hypocalcemia) or can reflect a real change in ionized [Ca^{2+}] (true hypocalcemia). False hypocalcemia is best excluded by direct measurement of ionized [Ca^{2+}]. Alternatively, total serum [Ca^{2+}] can be corrected for serum albumin as described previously. Causes of true hypocalcemia can be divided into conditions that are associated with high or low serum phosphate concentrations (Fig. 11.11).

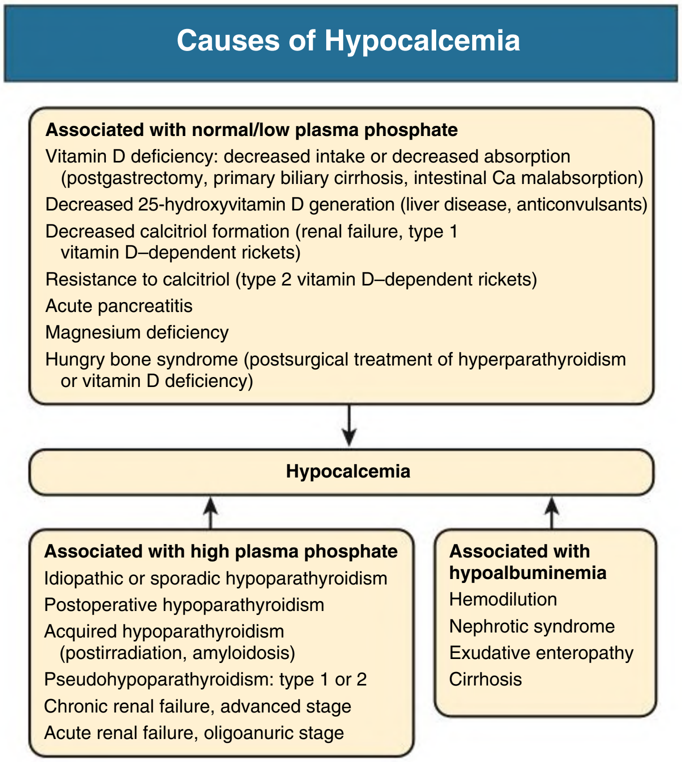

Fig. 11.11 Causes of hypocalcemia.

Hypocalcemia Associated With Hyperphosphatemia

Chronic and Acute Kidney Disease

A gradual loss of functioning nephrons leads to reduced calcitriol production and a subsequent decrease in intestinal calcium absorption. In parallel, serum phosphate concentrations progressively rise once the glomerular filtration rate (GFR) falls below approximately 35 mL/min/1.73 m^2. Kidney disease and hyperphosphatemia stimulate the release of fibroblast growth factor 23 (FGF-23), a bone-derived hormone that further suppresses calcitriol synthesis by inhibiting 1-alpha hydroxylase activity. However, hypocalcemia and hyperphosphatemia arising from CKD may be masked by treatments, including calcium-containing phosphate binders and vitamin D receptor agonists.

Hypocalcemia may also be observed in the setting of acute kidney injury (AKI). Several mechanisms may contribute to hypocalcemia in AKI, including an abrupt decline in calcitriol synthesis, an increase in FGF-23, and incomplete PTH response to the fall in serum [Ca^{2+}]. Hypocalcemia may also be observed in the polyureic recovery phase of AKI, especially after rhabdomyolysis.

Hypoparathyroidism

Under normal conditions, PTH increases serum [Ca^{2+}] and enhances urinary excretion of phosphate. Loss of PTH (hypoparathyroidism) therefore leads to hypocalcemia and hyperphosphatemia. Hypoparathyroidism may occur after intentional (parathyroidectomy) or nonintentional (thyroidectomy) removal of the parathyroid glands. Hypoparathyroidism can also be caused by autoimmune destruction of parathyroid tissue, radiation, infiltrative diseases, or genetic conditions affecting development and/or function of the parathyroid glands. Sporadic cases are occasionally seen in patients with pernicious anemia or adrenal insufficiency. Pseudohypoparathyroidism (iPPSD;

Inactivating PTH/PTHrP Signaling Disorders, including former pseudohypoparathyroidism type 1a and 1b) is a group of rare genetic diseases characterized by isolated or syndromic end organ resistance to PTH.[21] Magnesium deficiency impairs the PTH response to hypocalcemia, producing a state of relative hypoparathyroidism. In addition, massive oral phosphate administration, such as used in bowel preparations, can also cause hypocalcemia with hyperphosphatemia, often in association with AKI.[22]

Hypocalcemia Associated With Hypophosphatemia

Hypocalcemia with hypophosphatemia may occur in vitamin D–deficient states because of insufficient daylight exposure, dietary deficiency of vitamin D, decreased absorption after GI surgery, intestinal malabsorption syndromes (steatorrhea), or hepatobiliary disease (primary biliary cirrhosis). However, overt hypocalcemia caused by substrate 25-hydroxyvitamin D deficiency is uncommon because 1,25-dihydroxyvitamin D is tightly maintained at much lower concentrations. Vitamin D–dependent rickets are a rare group of inherited disorders caused by defects in vitamin D metabolic enzymes (discussed in the section on hypophosphatemia).

Hypocalcemia is observed in the majority of critically ill adults and children. No singular mechanism has been identified; defects in calcitriol synthesis, PTH release, and target organ sensitivity to these hormones have been described. Hypocalcemia is also seen with hypomagnesemia in patients with acute pancreatitis caused by saponification in necrotic fat.

An acute decrease in ionized [Ca^{2+}] occurs during acute hyperventilation and the respiratory alkalosis that follows, regardless of the cause of hyperventilation.

Clinical Manifestations

As with hypercalcemia, the symptoms of hypocalcemia depend on the rate of development and severity. The most common manifestations, in addition to fatigue and muscular weakness, are increased irritability, loss of memory, a state of confusion, hallucination, paranoia, and depression. The best-known clinical signs are the Chvostek sign (tapping of facial nerve branches leading to twitching of facial muscle) and the Trousseau sign (carpal spasm in response to forearm ischemia caused by inflation of a sphygmomanometer cuff). Patients with acute hypocalcemia may have paresthesias of the lips and the extremities, muscle cramps, and occasionally frank tetany, laryngeal stridor, or seizures. Chronic hypocalcemia may be associated with cataracts, brittle nails with transverse grooves, dry skin, and decreased or even absent axillary and pubic hair, especially in idiopathic hypoparathyroidism.

On the electrocardiogram (ECG), the corrected QT interval is frequently prolonged, and arrhythmias may occur. The electroencephalogram shows nonspecific signs such as an increase in slow, high-voltage waves. Intracranial calcifications, notably of the basal ganglia, are observed radiographically in 20% of patients with early-onset hypoparathyroidism but much less frequently in patients with postsurgical hypoparathyroidism or pseudohypoparathyroidism.

Treatment

Therapy of hypocalcemia is directed toward the underlying cause. Severe and symptomatic (tetany) hypocalcemia requires rapid treatment with calcium gluconate as an IV bolus (e.g., 10 mL 10% weight/volume [2.2 mmol of calcium], diluted in 50 mL of 5% dextrose in water or isotonic saline), followed by 12 to 24 g over 24 hours. Calcium gluconate is preferred to calcium chloride, which can lead to extensive skin necrosis in accidental extravasation. Acute respiratory alkalosis, if present, should be corrected if possible.

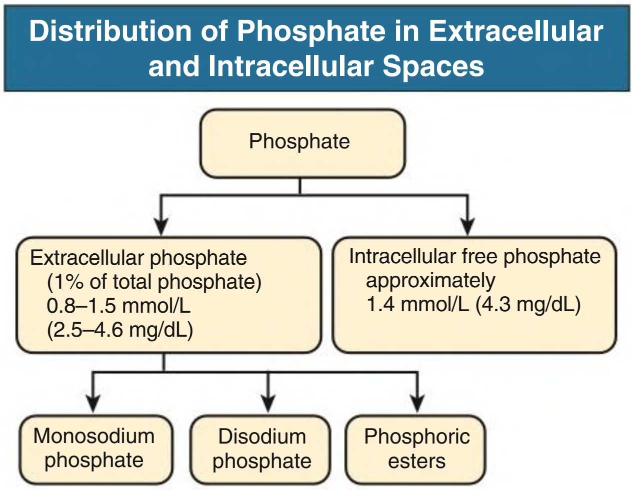

Fig. 11.12 Phosphate distribution in extracellular and intracellular spaces.

Treatment of chronic hypocalcemia includes oral administration of calcium salts, thiazide diuretics, and vitamin D. The amount of elemental calcium of the various salts differs greatly: the calcium content is 40% in carbonate, 36% in chloride, 12% in lactate, and only 8% in gluconate salts. The daily amount prescribed can be 1 to 4 g elemental calcium in adults. Concurrent magnesium deficiency (serum [Mg^{2+}] <0.70 mmol/L) should be treated with oral magnesium oxide or carbonate 100 to 300 mg elemental magnesium/day or with magnesium sulfate 4 to 8 mmol/day intramuscularly or 20 to 40 mmol/day in D5W IV.

Treatment of hypocalcemia secondary to hypoparathyroidism is difficult because urinary calcium excretion increases markedly with calcium and active vitamin D supplementation, potentially leading to nephrocalcinosis/nephrolithiasis and loss of kidney function. To reduce urinary calcium, thiazide diuretics can be used in association with restricted salt intake.

Therapy with vitamin D receptor agonists is the conventional treatment for idiopathic or acquired hypoparathyroidism because these compounds are better tolerated than massive doses of calcium salts. Large amounts of active vitamin D and calcium are often required for extended periods after surgical parathyroidectomy because of massive redistribution of calcium into the bone (hungry bone syndrome). Such patients require regular monitoring to avoid hypercalcemia, hyperphosphatemia, and hypercalciuria.

PHOSPHATE HOMEOSTASIS

Distribution of Phosphate in the Organism

Phosphorus is distributed in the body as 85% within bone and teeth; 15% inside cells; and less than 1% in the extracellular fluid, which includes the circulation. Fig. 11.12 shows the distribution of phosphate in extracellular and intracellular compartments. Within cells, phosphate regulates key enzymatic processes and serves as an essential structural component of nucleic acids and phospholipid membranes. In the bloodstream, phosphate circulates as HPO_4^{2-} and $H_2PO_4^-$ in an approximate 4 to 1 ratio at physiologic pH. Normal serum phosphate concentrations range from 2.8 to 4.5 mg/dL (0.9–1.5 mmol/L) with known circadian fluctuation.[23] The lowest concentrations are observed in the morning between 8 AM and 10 AM, and relatively higher concentrations are seen in late afternoon.[24] Dietary phosphate intake has only a minimal impact on the serum phosphate concentration in normal individuals and nondialyzed persons with mild to moderate CKD.

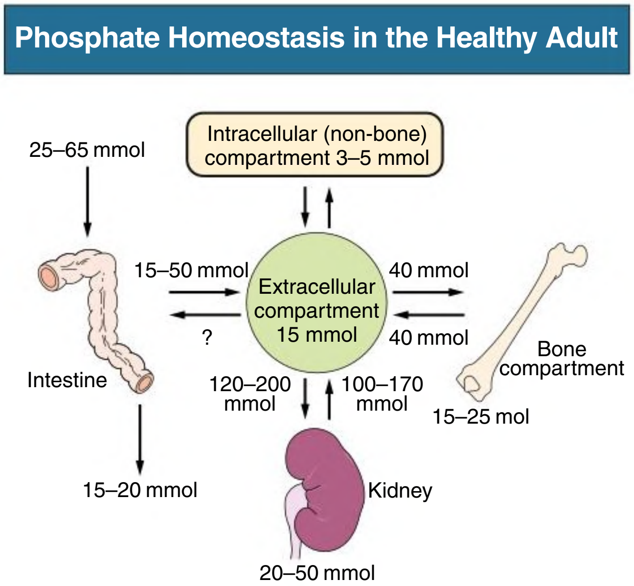

Fig. 11.13 **Phosphate Homeostasis in Healthy Adults.** At net zero balance, identical net intestinal uptake (absorption minus secretion) and urinary loss occur. After its passage into the extracellular fluid, phosphate enters the intracellular space, is deposited in bone or soft tissue, or is eliminated via the kidneys. Entry and exit fluxes between the extracellular and intracellular spaces (skeletal and nonskeletal compartments) are also the same under steady-state conditions.

Fig. 11.13 shows the balance of ingestion, body distribution, and excretion of phosphate in a healthy person. Phosphate needs are highest in children during growth. Phosphates are widely found in milk products, meat, eggs, and cereals, and are used extensively as food additives.

Phosphate enters epithelial cells via cotransport with sodium. Three different Na^+-Pi cotransporter families have been identified and characterized. Members of the type 1 Na^+-Pi family participate in kidney uric acid transport but do not directly impact phosphate homeostasis.[25] The members of the type 2 family of Na-Pi cotransporters are the key players in phosphate regulation and are NPT2a, NPT2b, and NPT2c (encoded by *SLC34A1*, *SLC34A2*, and *SLC34A3*). NPT2a and NPT2c are expressed in the apical brush border of the proximal tubules, whereas NPT2b is expressed in the small intestine.[26–29] The type 3 family of Na^+-Pi cotransporters (encoded by *SLC20A1* and *SLC20A2*) is ubiquitously expressed. Type 3 transporters do not participate directly in phosphate homeostasis; however, they mediate phosphate entry into vascular smooth muscle cells, which is hypothesized to be an important step in the pathogenesis of kidney-related vascular calcification.[30] Mutations in *SLC20A2* are the cause of primary familial brain calcification syndrome, a rare condition characterized by progressive bilateral calcification of the basal ganglia, thalami, and cerebellum.[31]

Intestinal, Renal, and Skeletal Handling of Phosphate

Intestinal phosphate absorption occurs via transepithelial and paracellular routes (Fig. 11.14). In contrast to the relatively restricted absorption of calcium, intestinal phosphate absorption is permissive, with 60% to 75% of dietary phosphate absorbed (15–50 mmol/day) in a linear, nonsaturable fashion (Fig. 11.15). The amount of intestinal phosphate absorption depends on the source of dietary phosphate: Absorption is highest for phosphate-containing food additives and lowest for plant-based phosphate sources.[32] Cations, such as calcium, magnesium, and aluminum, bind to phosphate in the intestinal tract, limiting its absorption.

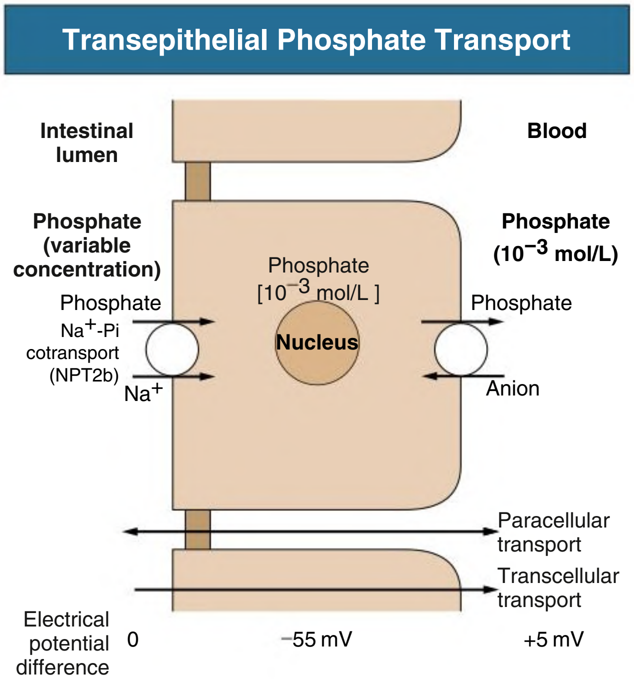

Fig. 11.14 **Transepithelial Phosphate Transport in the Small Intestine.** Phosphate enters the enterocyte (influx) through the brush border membrane using the Na^+/Pi cotransport system, with a stoichiometry of 2:1, operating against an electrochemical gradient. Phosphate exit at the basolateral side possibly occurs by passive diffusion or more probably by anion exchange.

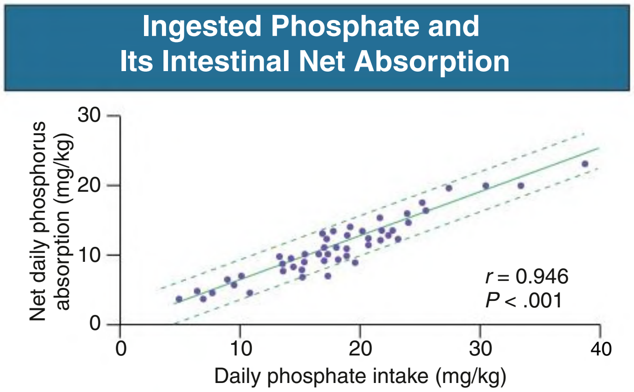

Fig. 11.15 **Phosphate Ingestion and Absorption.** Relationship between ingested phosphate and phosphorus absorbed in the digestive tract (net absorption) in healthy young adults. (From Wilkinson R. Absorption of calcium, phosphate, and magnesium. In: Nordin BEC, ed. *Calcium and Magnesium Metabolism.* Churchill Livingstone; 1976:36–112.)

Calcitriol is the primary hormonal determinant of intestinal phosphate absorption. Calcitriol upregulates NPT2b cotransporters in the small intestine, increasing phosphate entry via transcellular transport.[33,34] Vitamin D receptor agonists, including calcitriol and its analogs, enhance phosphate and calcium absorption, whereas nonactivated vitamin D supplements, such as cholecalciferol and ergocalciferol, have only a minimal impact on intestinal uptake. In experimental models, nicotinamide, a form of vitamin B_3, inhibits NPT2b expression, suggesting a role in the treatment of hyperphosphatemia. However, nicotinamide had no effect on serum phosphate concentrations in patients with moderate to advanced CKD in a randomized trial.[35]

Nomogram for Estimation of the Renal Threshold Phosphate Concentration

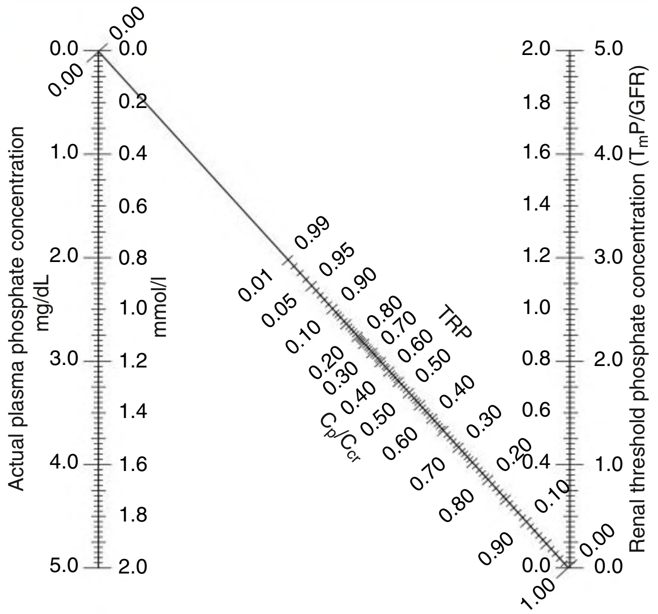

Fig. 11.16 Nomogram for Estimation of Kidney Threshold Phosphate Concentration. A straight line through the appropriate values of phosphate concentration and TRP (amount of phosphate reabsorbed, or C_P/C_{cr}, where *C* is clearance for phosphate or creatinine) passes through the corresponding value of T_mP/GFR. *GFR,* Glomerular filtration rate. (From Murer H, Hernando N, Forster I, Biber J. Proximal tubular phosphate reabsorption: Molecular mechanisms. *Physiol Rev.* 2000;80[4]:1373–1409.)

The kidneys play a major role in phosphate homeostasis by controlling its excretion. In both animals and humans, ingestion of a phosphate-containing meal leads to rapid excretion of phosphate in the urine without detectable changes in serum concentrations.[36] Phosphate is minimally protein bound, freely filtered in the glomerulus, and reabsorbed to a variable extent by the proximal tubules to match the body's needs. Under steady-state conditions, urinary phosphate excretion equals phosphate absorption through the gut. This balance is typically achieved by excretion of 5% to 20% of the filtered phosphate load; however, higher fractional excretions are usually needed to maintain balance in advanced CKD because of the progressive loss of filtering nephrons. The proportion of phosphate reabsorption can be estimated by the urinary fractional excretion of phosphate ($FEPO_4$).

$$\text{Fraction excretion of phosphate } (FEPO_4) = \frac{(\text{Urine phosphate}) \times (\text{Serum creatinine}) \times 100\%}{(\text{Serum phosphate}) \times (\text{Urine creatinine})}$$

$FEPO_4$ values are typically less than 20% in people without kidney disease and can reach values of 50% or higher in advanced CKD to compensate for the loss of functional nephrons.[37] An alternative method for assessing kidney phosphate reabsorption is the maximal tubular reabsorption of phosphate (TmP) factored for GFR (TmP/GFR, Bijvoet index). This value represents the concentration above which most phosphate is excreted in the urine and below which most is reabsorbed. TmP/GFR is calculated from the serum phosphate concentration and the tubular reabsorption of phosphate (Fig. 11.16).

After passage through the glomerulus, phosphate is reabsorbed via NPT2a and NPT2c cotransporters in the proximal tubule. The expression of these transporters, and thus total kidney phosphate reabsorption, is primarily regulated by PTH, FGF-23, and the serum phosphate concentration itself. PTH and FGF-23 downregulate NPT2a and NPTc on the apical brush border, thereby increasing urinary phosphate excretion.[38,39] PTH is believed to act by modulating the sodium-hydrogen exchanger regulator factor-1 (NHERF-1), a scaffolding protein that links membrane bound NPTa to the cytoskeleton.[40] FGF-23 is a hormone that is produced in bone and binds to target receptors in the kidneys. The transmembrane coreceptor α-klotho stabilizes the FGF-23-receptor interaction to promote downregulation of NPT2a and NPT2c on the apical cell surface.[41] Genetic disruption of FGF-23 or klotho in animal models results in an identical phenotype of hyperphosphatemia, calcitriol toxicity, vascular calcification, and premature death.[42]

Although PTH and FGF-23 act synergistically to enhance kidney phosphate excretion, these hormones have opposing actions on vitamin D metabolism. PTH promotes the synthesis of calcitriol by stimulating 1-alpha hydroxylase, which converts substrate 25-hydroxyvitamin D into its active form.[43] In contrast, FGF-23 inhibits calcitriol synthesis by suppressing 1-alpha hydroxylase and by catalyzing its conversion to an inactive 24-hydroxylated compound.[44]

Bone permanently exchanges phosphate with the surrounding milieu. Entry and exit of phosphate from bone amount to approximately 10 mmol/day (slowly exchangeable phosphate) for a total skeleton content of approximately 20,000 mmol. The net balance is positive during growth, zero in the young adult, and negative in older persons.

HYPERPHOSPHATEMIA

Causes of Hyperphosphatemia

Hyperphosphatemia is commonly caused by impaired phosphate excretion in acute or chronic kidney disease.[45] Hyperphosphatemia may also be caused by increased exogenous or endogenous phosphate supply (Fig. 11.17).

Acute Kidney Injury

An acute reduction in GFR leads to phosphate retention and a subsequent increase in serum phosphate. Serum concentrations may reach extremely high values if there is concomitant release from tissues, such as in rhabdomyolysis. Serum FGF-23 concentrations increase rapidly in the setting of AKI, demonstrating a unique sensitivity of this hormonal system to changes in kidney function, phosphate metabolism, or both signals in combination.[46]

Chronic Kidney Disease

Serum phosphate concentrations generally remain within the normal range in CKD until the GFR falls below about 35 mL/min per 1.73m^2.[47] The preservation of serum phosphate concentrations in CKD highlights the complex regulatory mechanisms involved in phosphate homeostasis. The gradual loss of filtering nephrons leads to phosphate retention, which stimulates the phosphaturic hormones PTH and FGF-23. These hormones defend the serum phosphate concentration by enhancing kidney phosphate excretion. However, the price paid for maintaining phosphate balance in CKD is chronic elevation of both PTH and FGF-23, which are associated with adverse clinical consequences, including bone and cardiovascular disease (see Chapters 85 and 88).[48,49] Moreover, FGF-23 potently suppresses calcitriol synthesis, possibly to limit further intestinal phosphate absorption, leading to reduced calcium absorption and ongoing stimulation of PTH. Eventually, the remaining filtering nephrons can no longer support further phosphate excretion, and serum concentrations begin to rise.

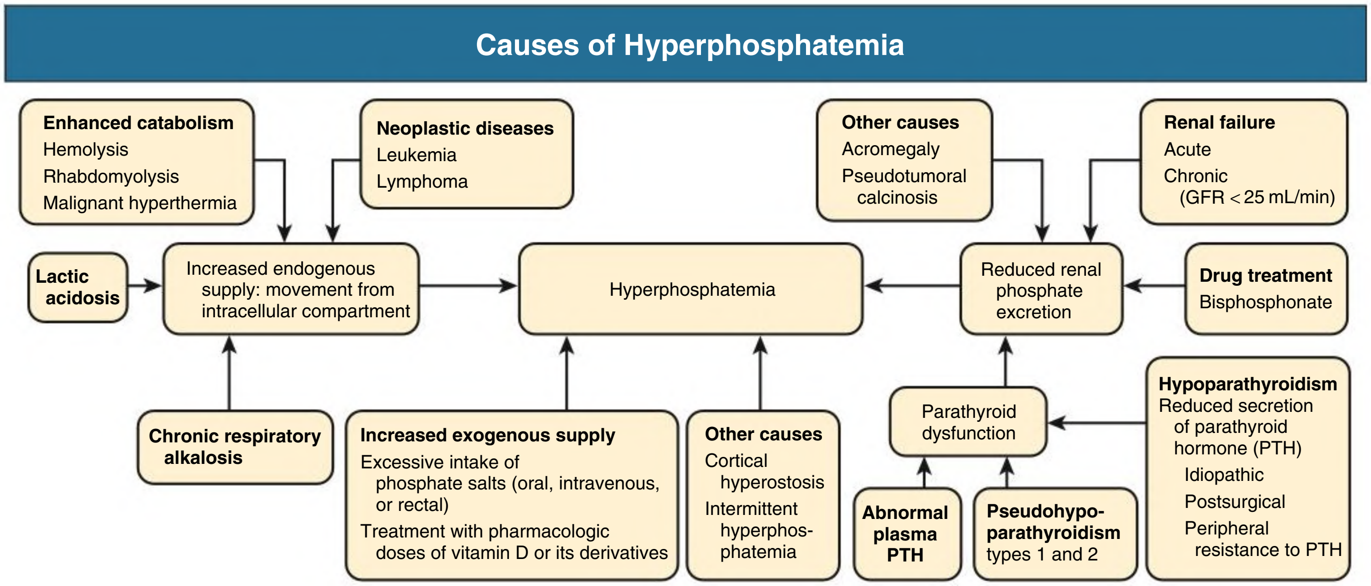

Fig. 11.17 Causes of hyperphosphatemia. *GFR*, Glomerular filtration rate; *PTH,* parathyroid hormone.

By this time, secondary hyperparathyroidism is typically evident, along with elevated FGF-23, calcitriol deficiency, and hypocalcemia.

Lytic States

Excess phosphate loss from tissues can be observed in states of extreme cell lysis, particularly rhabdomyolysis (crush injury) or malignant neoplasms and their treatment (especially lymphomas and leukemias). Hyperphosphatemia in rhabdomyolysis is typically accompanied by hypocalcemia, myoglobinuria, and AKI. Severe hypercatabolic states, such as sepsis or diabetic ketoacidosis, can also cause hyperphosphatemia by increasing phosphate release from cells (which is usually accompanied by an acute reduction in GFR).

Treatment-Induced Hyperphosphatemia

A massive supply of exogenous phosphate, as may occur with phosphate-based laxatives or enemas, can cause hyperphosphatemia. Oral sodium phosphate solutions for colonoscopy contain large quantities of phosphate that can cause precipitation of calcium phosphate crystals within the kidney tubules and AKI.[50] Recovery from this condition is slow and often incomplete, with some cases resulting in permanent dialysis. For these reasons, bowel preparations other than those based on sodium phosphate salts should be used in patients with CKD. Bisphosphonates, in particular etidronate in Paget disease, can sometimes increase serum phosphate concentrations, possibly through increased liberation of tissue phosphate or an increase in tubular reabsorption.

Hypoparathyroidism

States of reduced PTH secretion (idiopathic or postsurgical hypoparathyroidism) or resistance to its peripheral action (pseudohypoparathyroidism) lead to diminished tubular excretion of phosphate. The resulting hyperphosphatemia increases the ultrafiltered load, leading to regulation of serum phosphate concentrations at a new steady state.

Acromegaly

Insulin-like growth factor 1 (IGF-1) and growth hormone (GH) enhance phosphate reabsorption in the proximal tubules by increasing tubular Npt2a expression. Acromegaly, a hormonal disorder characterized by excess GH and stimulation of IGF-1, is frequently associated with hyperphosphatemia.[51]

Familial Tumoral Calcinosis

This rare autosomal recessive disorder is caused by mutations in either *GALNT3* or *FGF23* genes and seen primarily in people of Middle Eastern or African ancestry. *GALNT3* encodes a glycosyl transferase that catalyzes the O-glycosylation of FGF-23 to protect the molecule from enzymatic cleavage. Disruption *GALNT3* yields a less stable form of FGF-23 that is proteolytically degraded within the cell.[52] Direct mutations in *FGF-23* in tumoral calcinosis also yield a less stable form of the molecule, resulting in a shared phenotype.[53] The lack of functional FGF-23 leads to unopposed phosphate reabsorption by the proximal tubules and calcitriol excess, resulting in hyperphosphatemia, hypercalcemia, and soft tissue calcifications. Serum PTH concentrations may be normal or mildly suppressed.

Respiratory Alkalosis by Prolonged Hyperventilation

Respiratory alkalosis caused by prolonged hyperventilation is characterized by kidney resistance to the actions of PTH, hyperphosphatemia, and hypocalcemia. There may also be functional pseudohypoparathyroidism because kidney phosphate clearance is diminished, whereas serum PTH is normal, despite hypocalcemia. There is no decrease in urinary calcium excretion.

Clinical Manifestations

Acute and severe hyperphosphatemia can cause hypocalcemia by blocking calcitriol synthesis via inhibition of 1-alpha hydroxylase. Chronic hyperphosphatemia is suspected to play a causal role in the pathogenesis of kidney-related soft tissue calcification and CKD-mineral bone disorder (see Chapter 88). In extreme cases, hyperphosphatemia can induce tumor-like deposits of calcium phosphate in soft tissue (Teutschlander disease; Fig. 11.18), crystals within the kidney tubules, and vascular calcification within the arteries of the skin (calciphylaxis or calcific uremic arteriolopathy; see Chapter 91).

Treatment

The treatment of acute hyperphosphatemia is aimed at increasing phosphate excretion by the kidneys, either by IV fluids or by kidney replacement therapies in severe AKI. IV dextrose and insulin promote a shift of phosphate into cells, similar to effects on potassium. The treatment of chronic hyperphosphatemia in patients with advanced

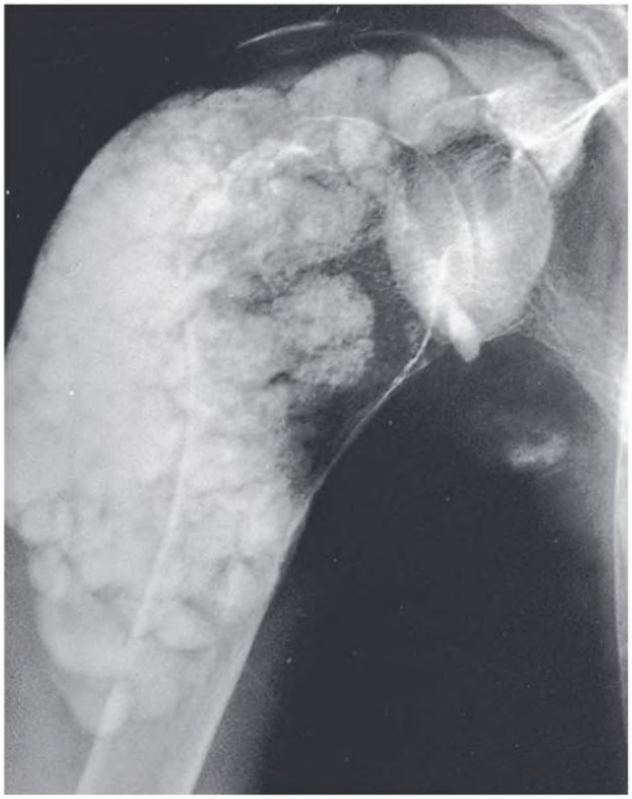

Fig. 11.18 Tumor-like extraskeletal calcification in the shoulder.

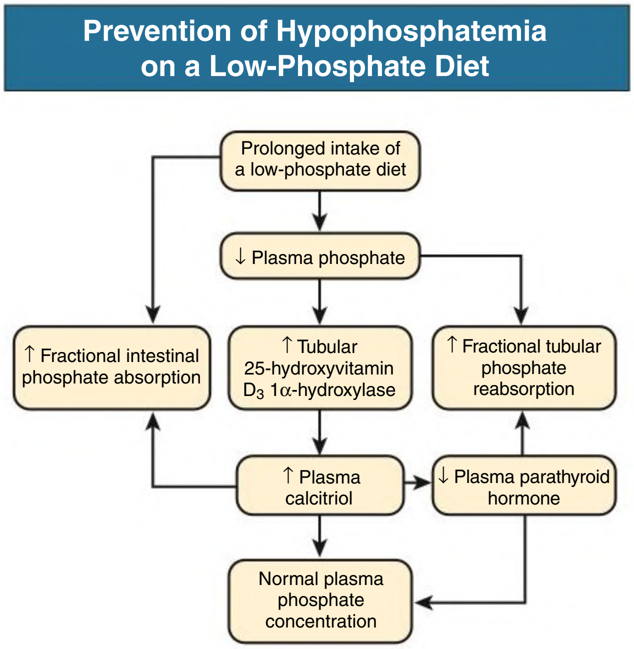

Fig. 11.19 **Prevention of Hypophosphatemia.** Compensatory mechanisms during prolonged intake of a phosphate-poor diet help prevent hypophosphatemia.

CKD and ESKD may require an oral phosphate binder, which complexes with phosphate in the GI tract to limit absorption (see Chapter 88). Phosphate is removed by dialysis; however, the rate of phosphate elimination during intermittent hemodialysis sessions declines over the course of a single treatment. More frequent sessions may be needed for efficient phosphate clearance.

HYPOPHOSPHATEMIA

Decreased serum phosphate concentrations may occur because of prolonged periods of reduced phosphate intake or malabsorption. However, as shown in Fig. 11.19, several defense mechanisms counter a reduction in serum phosphate that may result from reduced intake.

Causes of Hypophosphatemia

Hypophosphatemia may be caused by genetic diseases or acquired conditions (Fig. 11.20). These conditions may be mechanistically classified into those caused by kidney phosphate wasting, a shift of phosphate into cells, or protracted states of reduced phosphate intake. The urinary phosphate excretion can be used to distinguish among these mechanistic processes. Severe hypophosphatemia (<0.3 mmol/L) is almost always caused by acquired rather than inherited disorders.

Inherited Forms of Hypophosphatemia

Inherited diseases causing hypophosphatemia are usually diagnosed in childhood, when they present with rickets, osteomalacia, dental abscesses, and enthesopathy. These conditions include genetic mutations that culminate in an excess of circulating FGF-23 (hypophosphatemic rickets), disorders of proximal tubular reabsorption (Fanconi syndrome), or defects secondary to another genetically transmitted disease, mainly metabolic disorders or disturbances in the actions of vitamin D.

X-Linked hypophosphatemic rickets. Hypophosphatemic rickets are a group of genetic disorders characterized by a shared phenotype of excess circulating FGF-23, which leads to kidney phosphate wasting, suppressed calcitriol synthesis, and impaired bone mineralization. X-linked hypophosphatemic rickets typically presents in childhood with chronic hypophosphatemia, skeletal deformities, osteomalacia, and short stature; however, the short stature may not be recognized until later in the disease process.[54] The condition is caused by mutations in *PHEX* (phosphate-regulating endopeptidase on the X chromosome), which is believed to play a role in the proteolysis of FGF-23.[55] *PHEX* mutations cause release of excess intact FGF-23 into the circulation with consequent kidney phosphate wasting. PTH levels are typically normal or slightly elevated in response to secondary calcitriol deficiency and hypocalcemia. The alkaline phosphatase level is elevated.

Autosomal-dominant hypophosphatemic rickets. Children with this phosphate-wasting disorder present with skeletal defects, including bowing of the long bones, widening of costochondral joints, and dental abscesses. The condition is caused by activating mutations in *FGF-23.* In contrast to familial tumoral calcinosis, in which *FGF-23* mutations yield an unstable form of the molecule that is prone to proteolysis, mutations in autosomal-dominant hypophosphatemic rickets yield an aberrant form of FGF-23 that is resistant to proteolytic cleavage.[56]

Autosomal-recessive hypophosphatemic rickets. This disorder is caused by mutations in either *DMP1, ENPP1,* or *FAM20C,* which result in increased production of intact FGF-23.[57] *ENPP1* and *FAM20C* mutations are also associated with pathologic calcification of the medium and large arteries.

Fanconi syndrome and proximal renal tubular acidosis. Fanconi syndrome (see Chapter 50) is characterized by complex transport defects in the proximal tubules that lead to impaired reabsorption of glucose, amino acids, bicarbonate, and phosphate. Because more than 70% of the filtered phosphate load is typically reabsorbed in the proximal tubules, Fanconi syndrome is associated with kidney phosphate wasting and hypophosphatemia. Inherited causes of

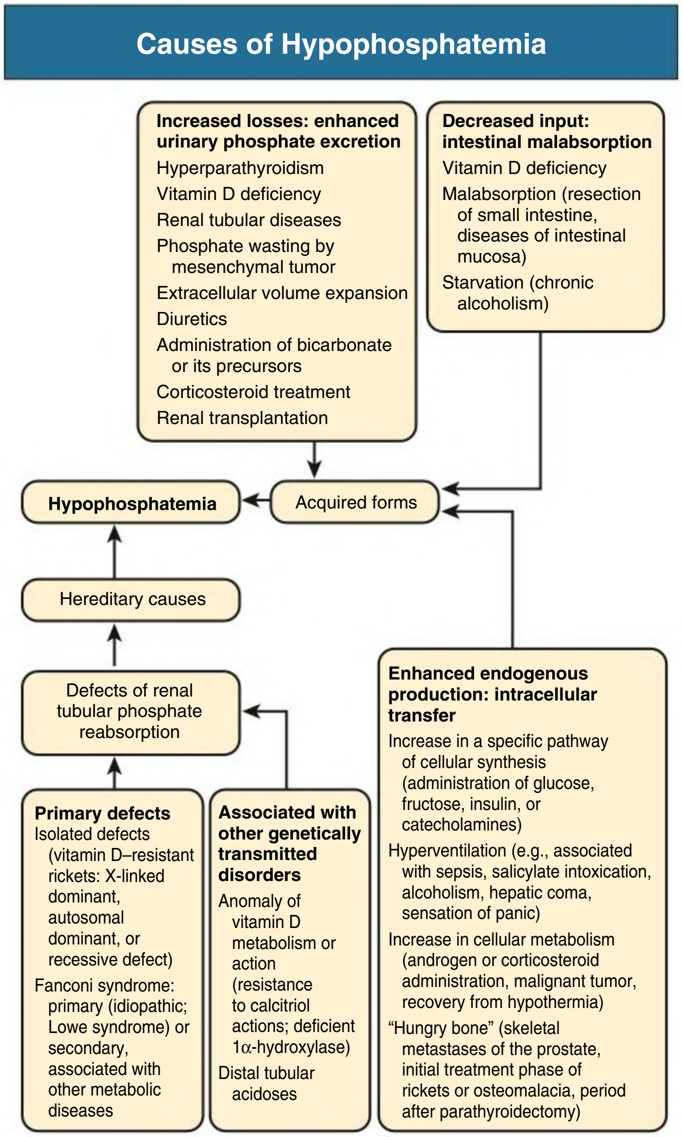

Fig. 11.20 Causes of hypophosphatemia.

Fanconi syndrome include primary disorders (Lowe syndrome, Dent disease) and other metabolic conditions (cystinosis, Wilson disease). In Dent disease and Lowe syndrome, defective recycling of megalin to the apical cell surface of the proximal tubule has been found, implicating a role in abnormal endocytic function.[58]

Fanconi syndrome with kidney phosphate wasting may also occur in adults as an acquired disorder. Common causes are multiple myeloma[59] and specific drugs, such as tenofovir, ifosfamide, and carbonic anhydrase inhibitors. In acquired Fanconi syndromes, kidney 1α-hydroxylase activity may be impaired, leading to reduced calcitriol synthesis and impaired bone mineralization.

Hypophosphatemia linked to other inherited diseases. Several rare inherited disorders of vitamin D metabolism may be associated with hypophosphatemia.[60] These include mutations in *CYP27B1,* which encodes 1α-hydroxylase, and *CYP2R1*. Inactivating mutations in *CYP27B1* (vitamin D–dependent rickets type I) typically present with severe hypocalcemia in the first year of life. The treatment is calcitriol replacement. Mutations in *VDR* (vitamin D–dependent rickets type II) also present early in life. In addition to hypocalcemia, hypophosphatemia, and bone disease, alopecia is a defining feature. *VDR* mutations may respond to extremely high doses of calcitriol depending on the severity of the mutation.

Distal renal tubular acidosis (type 1). Distal renal tubular acidosis type 1 (see Chapter 13) is associated with hypercalciuria and nephrocalcinosis because chronic metabolic acidosis enhances citrate reabsorption in the proximal tubules, preventing formation of soluble calcium-citrate complexes in the urine. Chronic acidosis also increases release of calcium and phosphate from bone. Hypophosphatemia is inconsistently found in this disorder.

Acquired Forms of Hypophosphatemia

The number of acquired diseases that are associated with hypophosphatemia is even greater than the inherited diseases (see Fig. 11.19). True phosphate deficiency associated with total body depletion must be distinguished from shifts of phosphate into cells or increased skeletal mineralization.

Alcoholism. Alcoholism is the most common cause of severe hypophosphatemia in Western countries. The mechanisms are multiple, including prolonged insufficient food intake, GI malabsorption, and, among hospitalized patents, glucose infusion, which stimulates insulin release driving phosphate from the extracellular to intracellular compartment.

Hyperparathyroidism. Primary hyperparathyroidism typically presents in middle age with asymptomatic hypercalcemia.[61] PTH downregulates NPT2a cotransporters in the proximal tubule causing kidney phosphate wasting and hypophosphatemia.

Posttransplantation hypophosphatemia. Kidney phosphate wasting is common in kidney transplant recipients, whether the organ is from a living or deceased donor. The primary cause is residual elevations of PTH, FGF-23, or both phosphaturic hormones in combination as a consequence of previous longstanding kidney disease.[62]

Acute respiratory alkalosis. In acute respiratory alkalosis, serum phosphate concentrations can sometimes decrease considerably to values as low as 0.1 mmol/L (0.3 mg/dL). Such a decrease is never observed in acute metabolic alkalosis. Hypophosphatemia that follows acute and intense hyperventilation is probably the result of muscle sequestration of extracellular phosphate. In contrast, prolonged chronic hyperventilation is associated with hyperphosphatemia (see previous discussion).

Diabetic ketoacidosis. During decompensated diabetes with acidosis provoked by the accumulation of ketone bodies, serum phosphate concentrations may initially be normal or high, even in the presence of hyperphosphaturia. Correction of ketoacidosis by insulin and refilling of the extracellular compartment leads to massive transfer of phosphate into the intracellular compartment, hypophosphatemia, and, subsequently, a reduction in urinary phosphate excretion. Serum phosphate concentrations typically do not decrease to less than 0.3 mmol/L (0.9 mg/dL) unless there is preexisting phosphate deficiency.

Total parenteral nutrition. Hyperalimentation can also be associated with severe hypophosphatemia because of insulin-mediated shift of phosphate into cells, particularly if phosphate is omitted from the parenteral nutrition solution. Severe hypophosphatemia is also a recognized complication of acute refeeding after starvation.

Tumor-induced osteomalacia. Certain mesenchymal tumors (hemangiopericytomas, fibromas, angiosarcomas) may express FGF-23 and other phosphatonins (sFRP-4, matrix extracellular phosphoglycoprotein [MEPE], or FGF-7). These tumors may present with a paraneoplastic syndrome of marked FGF-23 excess (tumor-induced osteomalacia) with clinical features of acute to subacute onset weakness, unexplained skeletal fractures, profound hypophosphatemia, and kidney phosphate wasting.[63] The condition resolves promptly after tumor resection.

Drug-induced hypophosphatemia. Hypophosphatemia may be caused by specific drugs. Tenofovir disoproxil fumarate (TDF) is associated with an acquired Fanconi syndrome that includes kidney phosphate wasting.[64] IV iron repletion with ferric carboxymaltose is associated with an abrupt rise in serum FGF-23, kidney phosphate wasting, and severe hypophosphatemia.[65] This drug is suspected to stabilize FGF-23 in the circulation. Imatinib mesylate, a tyrosine kinase inhibitor, and sorafenib, a vascular endothelial growth factor (VEGF) inhibitor, are both associated with hypophosphatemia[66]; the mechanism remains unclear.

Continuous kidney replacement therapies. Hypophosphatemia is commonly seen in patients receiving continuous venovenous hemofiltration or hemodialysis for AKI because of the constant removal of phosphate by these modalities. Phosphate repletion or the addition of phosphate to the replacement fluid is often required to maintain serum phosphate concentrations within the normal range.

Clinical Manifestations

Clinical manifestations of hypophosphatemia depend on the rate of onset more than the severity or total body phosphate deficit. In practice, hypophosphatemia is not clinically evident until serum phosphate concentrations are less than 0.65 mmol/L (2.0 mg/dL). Manifestations include metabolic encephalopathy, red and white blood cell dysfunction, sometimes hemolysis, and thrombocytopenia. Reduced muscle strength and decreased myocardial contractility (with occasional rhabdomyolysis and cardiomyopathy, respectively) may also occur.

Treatment

Hypophosphatemia is generally not an emergency. First, the mechanism involved should be defined to determine the most appropriate treatment. Inherited conditions characterized by FGF-23 excess (hypophosphatemic rickets) are treated by phosphate repletion and calcitriol replacement. A recombinant human immunoglobulin G1 (IgG1) monoclonal antibody to FGF-23 (Burosumab) has been developed and approved for the treatment of X-linked hypophosphatemic rickets.[67] This treatment has also been used in cases of tumor-induced osteomalacia in which the tumor cannot be surgically resected. For kidney transplant recipients with hypophosphatemia, treatment is directed toward refractory hyperparathyroidism, which is the usual underlying cause along with elevated FGF-23. Initial treatments include supplementation with cholecalciferol and, if serum phosphate concentrations are extremely low (<0.5 mmol/L), oral phosphate replacement. Management of refractory hyperparathyroidism in kidney transplantation shares similarities with primary hyperparathyroidism: Consideration for parathyroidectomy is based on the severity of hypercalcemia and accompanying bone and other complications of the disease. Cinacalcet may be considered for nonsurgical candidates.

When phosphate deficiency is diagnosed, oral treatment by milk products or phosphate salts should be tried first whenever possible, except in the presence of nephrocalcinosis or nephrolithiasis with urinary phosphate wasting. In severe symptomatic deficiency, phosphate can also be infused by IV in divided doses over 24 hours. In patients undergoing parenteral nutrition, 10 to 25 mmol potassium phosphate should be given for each 1000 kcal, with care taken to avoid hyperphosphatemia because of the risk of inducing soft tissue calcification. Dipyridamole (300 mg divided into four doses per day) reduces the urinary excretion of phosphate in patients with a low renal phosphate threshold.

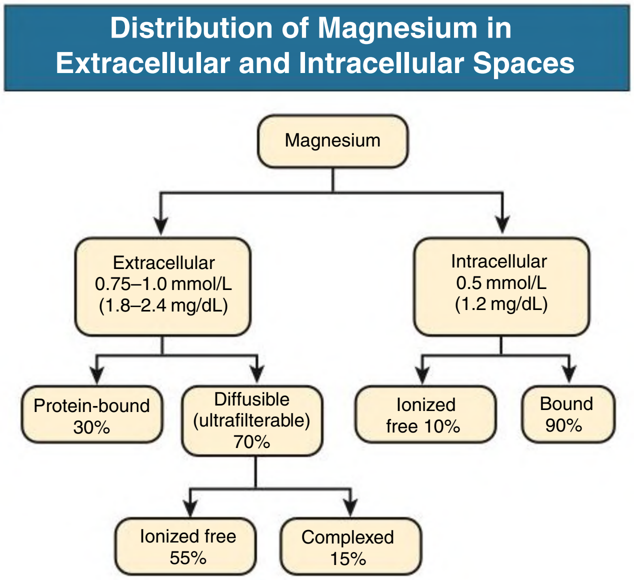

Fig. 11.21 Magnesium Distribution in Extracellular and Intracellular Spaces. Intracellular magnesium concentration is that of free, not total, magnesium.

MAGNESIUM HOMEOSTASIS AND DISORDERS OF MAGNESIUM METABOLISM

Distribution of Magnesium in the Organism and Magnesium Homeostasis

Magnesium is the second most abundant cation in the intracellular fluid after potassium. Magnesium contributes to the regulation of mitochondrial function, inflammatory processes, growth, neuronal activity, cardiac excitability, neuromuscular transmission, vasomotor tone, and blood pressure. The distribution of magnesium within the intracellular and extracellular spaces is shown in Fig. 11.21. Similar to calcium and phosphate, circulating magnesium concentrations [Mg^{2+}] represent only a minute fraction of total body magnesium.

Fig. 11.22 shows the balance of ingestion, body distribution, and excretion of magnesium in healthy individuals. Unlike calcium and phosphate, there is no hormonal system known to specifically regulate whole-body magnesium homeostasis. Magnesium moves into and out of cells through active transport systems. Insulin promotes magnesium influx into cells, whereas stimulation of β-adrenoceptors favors magnesium efflux.[68,69]

Intestinal and Kidney Handling of Magnesium

Magnesium is present in a wide variety of foods, especially nuts, seeds, and beans. Sufficient dietary intake is about 300 mg/day (12 mmol/day). Intestinal magnesium absorption varies from 25% to 60% with a mean of approximately 30%. Compared with calcium and phosphate, magnesium absorption occurs more distally in the GI tract, including the colon. Transcellular magnesium transport in the intestine is mediated by the melastatin-related transient receptor potential cation channels 6 and 7 (TRPM6 and TRPM7).[70,71]

The kidneys are the primary route of magnesium elimination because losses through intestinal secretion and sweat are small under normal conditions. Approximately 70% of circulating [Mg^{2+}] is filtered in the glomerulus (104 mmol or 2500 mg/day). Tubular reabsorption occurs primarily in the thick ascending loop of Henle (65%) with additional reabsorption in the proximal tubules (25%) and distal convoluted tubules (5%; Fig. 11.23).[72] Under normal conditions, the

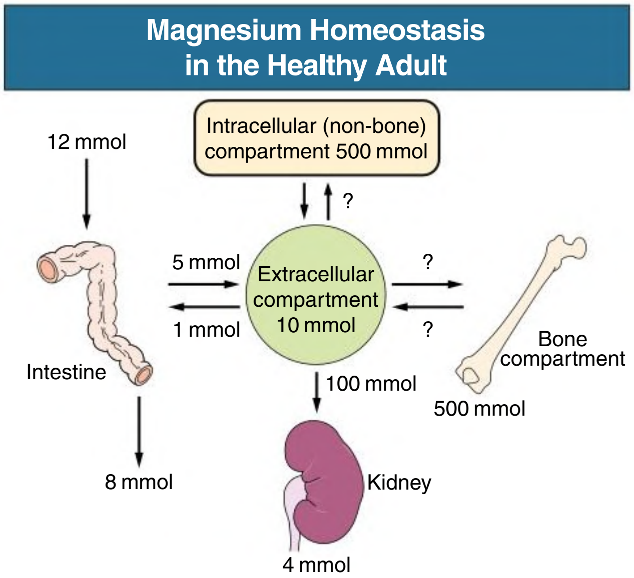

Fig. 11.22 **Magnesium Homeostasis in Healthy Adults.** Net zero balance results from net intestinal uptake (absorption minus secretion) equaling urinary loss. After its passage into the extracellular fluid, Mg^{2+} enters the intracellular space, is deposited in bone or soft tissue, or is eliminated via the kidneys. Entry and exit fluxes between the extracellular and intracellular spaces (skeletal and nonskeletal compartments) are also of identical magnitude; however, precise values of exchange are still debated.

urinary excretion of magnesium represents approximately 5% of the filtered load (4–5 mmol or 100 mg/day). This value constitutes a general cut point for assessing conditions of kidney magnesium wasting; a fractional magnesium excretion greater than 5% in the presence of hypomagnesemia suggests inappropriate urinary losses.

The bulk reabsorption of salt and water in the proximal tubules drives parallel reabsorption of chloride, potassium, and calcium. Yet the proximal tubules are relatively impermeable to magnesium accounting for its proportionately small reabsorption in this segment.[73] Members of the claudin family of proteins are suspected to facilitate magnesium transport in the proximal tubules; however, the specific components remain unknown.

In the TAL, magnesium is transported via paracellular claudin channels in tight junctions. Specifically, claudins 16 and 19 are hypothesized to include a cation-selective pore that facilitates calcium and magnesium reabsorption.[74] This transport is driven by reabsorption of Na^+, K^+, and $2Cl^-$ through the apical NKCC2 cotransporter, which generates an electrical lumen-positive gradient (see Fig. 11.9). The renal outer medullary potassium (ROMK) channel promotes NKCC2 activity by recycling potassium into the lumen, thereby enhancing magnesium reabsorption. Stimulation of the basolateral CaSR by calcium or magnesium directly decreases paracellular permeability, reducing magnesium reabsorption. Defects in NKCC2 caused by genetic disorders, specific drugs, or electrolyte imbalances lead to kidney magnesium wasting (see later discussion).[28] Approximately 5% of the filtered Mg^{2+} load is reabsorbed in the distal convoluted tubule

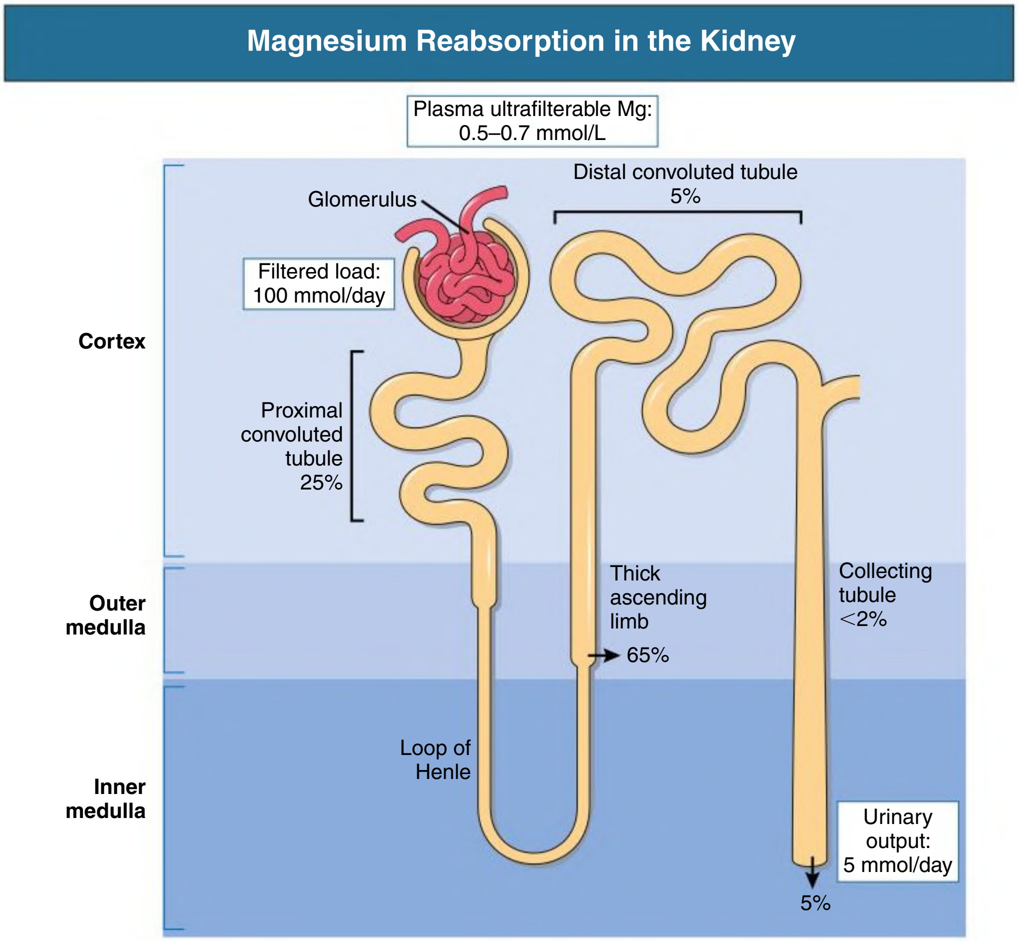

Fig. 11.23 **Sites of Magnesium Reabsorption in Various Segments.** Percentage absorbed in various segments of the renal tubule from the glomerular ultrafiltrate. (Redrawn from Shimada T, Kakitani M, Yamazaki Y, et al. Targeted ablation of Fgf23 demonstrates an essential physiologic role of FGF23 in phosphate and vitamin D metabolism. *J Clin Invest.* 2004;113[4]:561–568.)

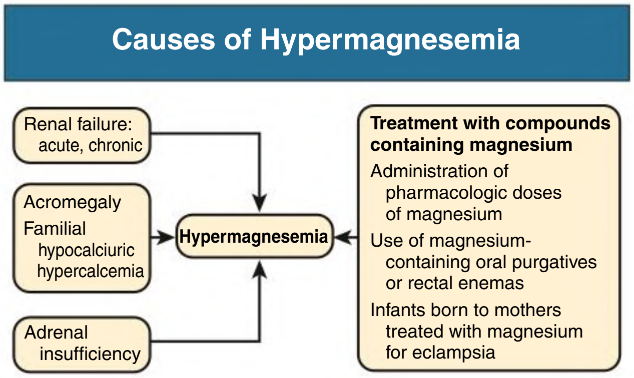

Fig. 11.24 Causes of hypermagnesemia.

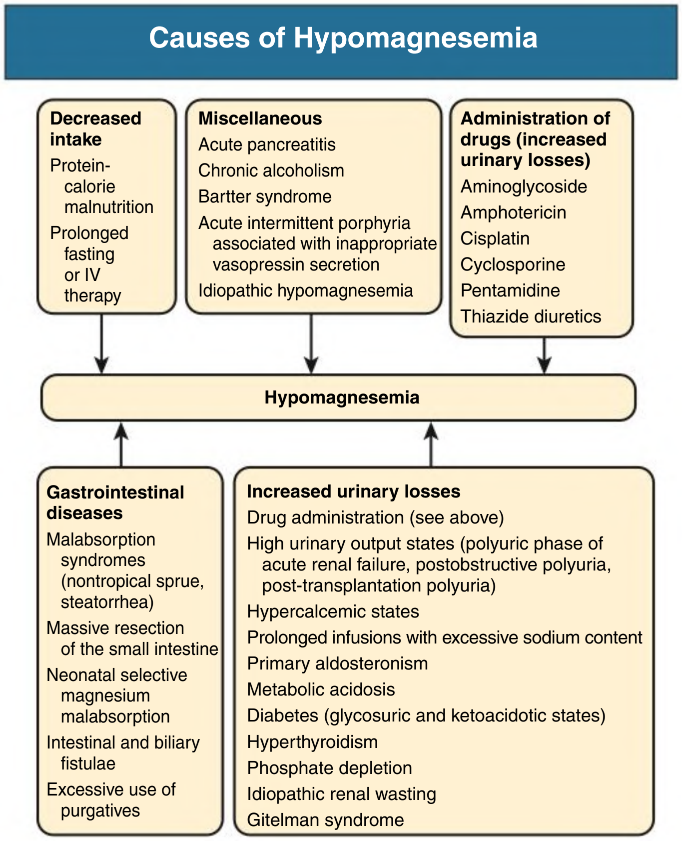

Fig. 11.25 Causes of hypomagnesemia.

via the identical TRPM6 channel found in the intestine. This channel colocalizes with the thiazide-sensitive Na^+ Cl^- cotransporter (NCC).[75] Conditions that impair NCC activity lead to reductions in magnesium reabsorption in the distal tubule.

Tubular magnesium transport is modulated by circulating [Mg^{2+}] and [Ca^{2+}] and by extracellular fluid volume. An increase in serum [Mg^{2+}] or [Ca^{2+}] reduces kidney magnesium reabsorption.[76] Extracellular volume expansion suppresses the reabsorption of sodium, chloride, potassium, and, to a lesser extent, magnesium in the proximal tubules.[77] Other hormones that can increase magnesium reabsorption include PTH, vasopressin, calcitonin, and glucagon.[78]

Hypermagnesemia

Elevated serum [Mg^2] can be seen in patients with acute kidney disease and CKD because of decreased elimination; however, enhanced single-nephron magnesium excretion in CKD tends to prevent overt hypermagnesemia. Elevated serum [Mg^{2+}] may also be observed with an increase in magnesium intake or administration, such as pharmacologic doses of magnesium for treating preeclampsia, oral magnesium-containing laxatives or rectal enemas, and Epsom salts (Fig. 11.24).[79] Mild hypermagnesemia may also occur in patients with adrenal insufficiency, acromegaly, and familial hypocalciuric hypercalcemia.

Clinical Manifestations

Symptoms and signs of hypermagnesemia are the result of the pharmacologic effects of magnesium on the nervous and cardiovascular systems. At concentrations of up to 1.5 mmol/L (3.6 mg/dL), hypermagnesemia is usually asymptomatic. Deep tendon reflexes are typically reduced or lost when serum [Mg^{2+}] exceeds 3.0 mmol/L (7.2 mg/dL). Respiratory paralysis, hypotension, abnormal cardiac conduction, and loss of consciousness may occur as serum [Mg^{2+}] approaches 5.0 mmol/L (12 mg/dL).

Treatment

Treatment of hypermagnesemia consists of cessation of magnesium administration and IV infusion of isotonic fluids and calcium salts, which promote kidney excretion. For management of symptomatic hypermagnesemia, calcium gluconate may be administered by IV as 1 g in 10 mL over 5 to 10 minutes (each gram of calcium gluconate is equal to approximately 90 mg of elemental calcium). Loop diuretics may also be used, where appropriate, to enhance kidney magnesium elimination.

HYPOMAGNESEMIA AND MAGNESIUM DEFICIENCY

Magnesium deficiency refers to a deficit in total body magnesium stores, whereas hypomagnesemia refers to low circulating magnesium concentrations. Although serum concentrations do not directly reflect total body magnesium stores, there is no practical test for whole-body deficiency. Therefore, clinical focus relates to hypomagnesemia. Causes can be broadly classified into conditions of reduced GI intake and disorders of kidney wasting (Fig. 11.25). The fractional excretion of magnesium can be used to distinguish between these mechanistic processes.

Dietary and Gastrointestinal Causes

A temporary reduction in dietary magnesium intake does not usually cause magnesium deficiency because of the remarkable ability of the kidneys to conserve magnesium. However, prolonged and severe dietary deficiency of less than 0.5 mmol/day can produce symptomatic deficiency. Chronic alcoholism is among the most common causes of hypomagnesemia in hospitalized patients predominantly because of marked nutritional deficiency and some degree of inappropriate kidney wasting.[69] Decreased intestinal absorption may also be seen in malabsorption syndromes, such as nontropical sprue, diarrheal illnesses, and after bowel resection. Proton pump inhibitors can inhibit intestinal magnesium absorption; chronic use of these medications is associated with hypomagnesemia.[80] Hypomagnesemia and hypocalcemia are also observed in patients with acute pancreatitis because of saponification in necrotic fat.

Kidney Causes

Diuretics are the most common acquired cause of kidney magnesium wasting in adults. Loop diuretics block the NKCC2 cotransporter, reducing the driving force for adjacent paracellular reabsorption. Thiazide diuretics block the NCC cotransporter in the distal tubule, which colocalizes with TRPM6, leading to reduced kidney magnesium reabsorption. Several other medications interfere with kidney magnesium reabsorption and are associated with hypomagnesemia: gentamicin, cisplatin, amphotericin, calcineurin inhibitors (cyclosporine and tacrolimus), and monoclonal antibodies to the epidermal growth factor receptor (cetuximab and panitumumab).[81] Hypercalcemia can also induce kidney magnesium wasting by activating the CaSR in the thick ascending limb.

Primary inherited conditions causing kidney magnesium wasting are Bartter and Gitelman syndromes. Bartter syndrome is an autosomal recessive disorder caused by mutations in genes encoding major ion transporters in the TAL.[82] These include the basolateral Cl^- channel (encoded by *CLCNKB*), the NKCC2 cotransporter (encoded by *SLC12A1*), and the ROMK channel (encoded by *KCNJ1*). Genetically mediated dysfunction of this kidney segment leads to a "loop diuretic-like effect" with loss of potassium, calcium, and magnesium in the urine. Bartter syndrome typically presents in childhood with features of growth retardation, polyuria, metabolic alkalosis, hypokalemia, and hypomagnesemia.

Gitelman syndrome is an autosomal recessive disorder caused by inactivating mutations in the gene encoding the NaCl transporter NCC (*SLC12A3*).[83] Genetic disruption of NCC produces a thiazide diuretic–like effect with loss of sodium, potassium, and magnesium in the urine but with increased reabsorption of calcium. In contrast to Bartter syndrome, Gitelman syndrome typically presents in adulthood with muscle cramps, fatigue, hypokalemia, and hypomagnesemia. Bartter and Gitelman syndromes can be distinguished by the kidney handling of calcium. By mimicking the action of loop diuretics, Bartter syndrome is associated with kidney calcium wasting. In contrast, by mimicking the action of thiazide diuretics, Gitelman syndrome is associated with enhanced tubular calcium reabsorption.

The presence of Bartter or Gitelman syndrome is suggested by a constellation of chronic metabolic alkalosis, hypokalemia, and hypomagnesemia in the context of normal to low blood pressures and absence of known diuretic treatment. Two conditions that share these metabolic features are volume depletion caused by vomiting and the surreptitious use of diuretics. The urinary chloride concentration can be helpful for excluding vomiting in this situation (urinary chloride concentration typically <20 mEq/L in states of volume depletion). Differentiating covert diuretic use from Bartter or Gitelman syndromes is difficult because these genetic conditions mimic the effects of loop and thiazide diuretics, respectively.

Clinical Manifestations

Specific clinical manifestations of hypomagnesemia may be difficult to appreciate because of concomitant hypocalcemia and hypokalemia. Moreover, moderate degrees of magnesium deficiency can be difficult to detect because clinical manifestations may be absent, and blood levels may not reflect total body magnesium. Clinical manifestations of moderate to severe magnesium depletion include generalized weakness and neuromuscular hyperexcitability with hyperreflexia, carpopedal spasm, seizure, tremor, and, rarely, tetany. Cardiac findings may include a prolonged QT interval, ST depression, torsades de pointes, and potentiation of digoxin toxicity. The role of magnesium deficiency in seizure activity is demonstrated by its effectiveness as a treatment; among women with eclampsia, IV magnesium administration is more effective than phenytoin for the treatment of seizures.[84]

An important physiologic consequence of hypomagnesemia is blunting of the PTH response to calcium, leading to a state of relative hypoparathyroidism. Reduced PTH secretion consequently results in hypocalcemia and reduced calcitriol synthesis.

Treatment

Magnesium deficiency is managed by administration of magnesium salts. Magnesium sulfate is generally used for parenteral therapy (1500–3000 mg [150–300 mg elemental magnesium] per day). A variety of magnesium salts are available for oral administration, including oxide, hydroxide, sulfate, lactate, chloride, carbonate, and pidolate. The use of oral magnesium is often limited by GI side effects, particularly diarrhea, at higher dosages.

In Bartter and Gitelman syndromes, magnesium and potassium supplementation along with increased sodium intake are initially used to address the attendant electrolyte disturbances. Amiloride, which blocks distal sodium and potassium exchange, can help restore serum potassium and magnesium concentrations if concomitant volume contraction can be avoided. The defective loop sodium transport in Bartter syndrome stimulates prostaglandin synthesis and activates the renin-angiotensin-aldosterone system. These metabolic disturbances are countered by nonsteroidal anti-inflammatory agents and angiotensin-converting enzyme inhibitors, respectively, promoting their use in Bartter syndrome with careful monitoring of kidney function.

SELF-ASSESSMENT QUESTIONS

1. Which is the most common cause of hypercalcemia in patients with a low serum PTH level?
   A. Secondary hyperparathyroidism
   B. Cholecalciferol or ergocalciferol therapy
   C. Malignant neoplasias
   D. Familial hypocalciuric hypercalcemia
   E. Hypomagnesemia
2. Which is a primary cause of hypocalcemia in patients with chronic kidney disease?
   A. Primary hypoparathyroidism
   B. Secondary hyperparathyroidism
   C. Impaired calcitriol synthesis
   D. Blunted response to FGF-23
   E. Treatment with phosphate binders
3. Which kidney tubular segment is responsible for regulating phosphate balance?
   A. Proximal tubule
   B. Thick ascending limb
   C. Distal tubule
   D. Outer collecting duct
   E. Inner collecting duct
4. Hypophosphatemic rickets describe a group of genetic conditions characterized by an increase in circulating concentrations of which molecule?
   A. Calcitriol
   B. Parathyroid hormone
   C. Calcium
   D. FGF-23
   E. Phosphate-regulating endopeptidase on the X chromosome
5. Which of the following is most helpful in distinguishing a genetic cause of hypomagnesemia from surreptitious vomiting?
   A. Serum calcium
   B. Serum phosphate
   C. Urinary chloride
   D. Urinary potassium
   E. Prolonged QT interval

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